

Evaluating early-warning indicators of critical transitions in natural aquatic ecosystems

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Ecosystems can show sudden and persistent changes in state despite only incremental changes in drivers. Such critical transitions are difficult to predict, because the state of the system often shows little change before the transition. Early-warning indicators (EWIs) are hypothesized to signal the loss of system resilience and have been shown to precede critical transitions in theoretical models, paleo-climate time series, and in laboratory as well as whole lake experiments. The generalizability of EWIs for detecting critical transitions in empirical time series of natural aquatic ecosystems remains largely untested, however. Here we assessed four commonly used EWIs on long-term datasets of five freshwater ecosystems that have experienced sudden, persistent transitions and for which the relevant ecological mechanisms and drivers are well understood. These case studies were categorized by three mechanisms that can generate critical transitions between alternative states: competition, trophic cascade, and intraguild predation. Although EWIs could be detected in most of the case studies, agreement among the four indicators was low. In some cases, EWIs were detected considerably ahead of the transition. Nonetheless, our results show that at present, EWIs do not provide reliable and consistent signals of impending critical transitions despite using some of the best routinely monitored freshwater ecosystems. Our analysis strongly suggests that a priori knowledge of the underlying mechanisms driving ecosystem transitions is necessary to identify relevant state variables for successfully monitoring EWIs.

competition | intraguild predation | trophic cascade | time series | resilience indicators

Ecosystems can exhibit multistate stability and occasionally sudden transitions from one regime to another despite only incremental changes in drivers (1–5). These critical transitions are characterized by the occurrence of alternative regimes under the same environmental conditions and by abrupt, discontinuous transitions between regimes when a critical threshold is exceeded. As each regime is stabilized by feedback loops, the thresholds for the forward and backward shifts may differ, resulting in hysteresis (5). A well-known example of this phenomenon is the nutrient-driven shift between a clear, macrophyte-dominated regime and a turbid, phytoplankton-dominated regime in some shallow lakes (6).

At the ecosystem level, such fundamental reorganizations affect ecological processes and hence ecosystem services, potentially incurring large economic costs. Therefore, reliable tools for assessing ecosystem resilience are sought, ideally providing management with time to avert an impending critical transition (7). Stabilization through feedback loops often precludes a systematic response in state variables ahead of the shift, hindering the prediction of critical transitions (8). The development of early-warning indicators (EWIs)

derived from bifurcation theory has drawn considerable interest for their ability to detect critical transitions, particularly given their promise of generalizability. These EWIs have been shown to precede critical transitions in modeled time series (8–10), experimental time series (11–13), reconstructed paleo-climate records (14, 15) and whole-lake experiments (16). Until now, however, an assessment of the generality and the detection power of EWIs on long-term monitoring data in aquatic systems is lacking (17).

EWIs are statistical metrics that quantify the loss of temporal or spatial resilience and thereby provide advance warning of the potential proximity to a critical threshold (18). Several EWIs are related to critical slowing down, a characteristic property of dynamic systems close to catastrophic local bifurcations (19). A bifurcation marks a threshold at which the stabilizing properties of the state of the system change. As the system approaches such a threshold, the return rate to equilibrium after a small perturbation slows, so that the system tends to become more similar to its own past, resulting in an increase in autocorrelation at lag-1 (AR-1) (20). This lack of decay of the impact of past perturbations also leads to a buildup in variance, typically measured as an increasing

Significance

Early-warning indicators (EWIs), statistical metrics of system resilience, have been hypothesized to provide advance warning of sudden shifts in ecosystems, or so-called “regime shifts.” Here we tested this hypothesis for four commonly used EWIs. We used empirical time series from five freshwater ecosystems with documented sudden, persistent transitions hypothesized to represent critical transitions. EWIs were detected in several of these long-term records, and in some cases several years before the transition; however, these EWIs varied in reliability, and agreement between indicators was low. Moreover, their applicability was strongly limited by the requirement for ecosystem-specific knowledge of transition-generating mechanisms and their drivers to choose relevant state variables for analysis.

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trend in SD (21). Concurrent increases in SD and AR-1 in a time series also produces greater variability in low-frequency processes compared with high-frequency processes in the power spectrum of a time series, which can be quantified as an increasing density ratio (DR) of variance at low to high frequencies (7, 22). In addition to critical slowing down, the system tends to remain longer at the basin boundary between the two alternative attractors, resulting in a skewed distribution of the state variable (SK) (23). The expected trends in AR-1, SD, DR, and SK are not exclusive indicators of critical transitions as false positives and false negatives can occur (24). Thus, before the application of EWIs based on critical slowing down, careful analysis of whether a system is actually undergoing a critical transition is required (25).

Abrupt changes in the state of an ecosystem can develop from several mechanisms, including (i) linear tracking of large changes in environmental conditions, (ii) nonlinear but continuous (reversible) responses to gradual changes in environmental conditions, and (iii) nonlinear discontinuous (irreversible) responses to gradual changes in environmental conditions (26). Whereas the first mechanism is distinguished by concurrent large changes in environmental drivers (e.g., a sudden increase in temperature), the difference between the other two mechanisms becomes apparent only when the driver is reversed. Thus, single step changes in time series cannot provide direct evidence of a critical transition (27). Given the rarity of empirical time series covering forward and backward shifts, identification of critical transitions in natural systems is difficult; however, by linking observed step changes to a mechanistic understanding of the driving processes that can give rise to bistability, one can hypothesize (but not prove) the existence of a potential critical transition (5, 28).

Lakes have been proposed as particularly suitable ecosystems to test for EWIs associated with critical transitions. The modular nature of lakes also allows for comparisons across different lakes (29). In aquatic systems, various ecological mechanisms have been shown to generate critical transitions between alternative states; the most commonly identified mechanisms include (i) competition between two or more species (2, 28); (ii) trophic cascades through inclusion or exclusion of top predators (16) or parasites (30), resulting in overexploitation traps; and (iii) intraguild predation (IGP) through resource competitors that also prey on one another (31, 32). In the present work, we selected 14 state variables of five well-documented case studies of freshwater critical transitions to test whether four commonly used EWIs (AR1, SD, SK, and DR) can be detected reliably in advance of the transition. We assessed how often these EWIs showed the same trends (agreement), and whether their behavior depended on the mechanism, type of state variable, magnitude of the step, or sampling frequency. Finally, we also tested how many years ahead of the transition EWIs were detectable. We strive to provide a comprehensive assessment of EWIs in some of the best-documented aquatic time series collected following standard monitoring schemes.

Results

Selection of Case Study Ecosystems and State Variables. We selected five case study ecosystems based on expert knowledge of well-described regime shifts that can be qualified as critical transitions in aquatic ecosystems: Lake Müggelsee (Germany; LMS), Lake Veluwemeer (The Netherlands; LVM), Lake Zwemlust (The Netherlands; LZL), Lake Washington (United States; LW), and Lake Võrtsjärvi (Estonia; LV). Based on the literature on these case study ecosystems, we identified relevant critical transition-generating mechanisms, which guided our choice of 14 state variables for EWI analysis. The case studies, mechanisms, state variables, and drivers are summarized in Table 1. A more detailed description of each case study and rationale for choosing the 14 state variables is presented in *SI Appendix, section S1*.

Transition Detection and Seasonal Adjustment. We used three complementary methods—piecewise linear regression, Pettit, and STARS (*Methods*)—to robustly assess the timing of transitions (i.e., breakpoints, defined as large, persistent step changes) in the time series of each state variable. We found one breakpoint in 12 state variables and two breakpoints in two state variables (Table 2 and Fig. 1), resulting in 16 prebreakpoint time series (i.e., from start to step change) for further analysis. In all state variables, the timing of the breakpoints confirmed the timing of transitions reported in the literature. Information on the data structure, including time period, sampling interval, number of data points, and the percentage of missing values in the prebreakpoint time series, are summarized in *SI Appendix, Table S2*. Each state variable showed significant differences in the means before and after the breakpoint (Table 2), with step magnitudes ranging from 0.24–1.59*standard deviation (Table 2). Each prebreakpoint time series was detrended and seasonally adjusted using a Gaussian smoother with a bandwidth corresponding to 12 data points (for monthly datasets) or 26 data points (for fortnightly datasets). Testing of the residual time series for remaining linear trends and seasonality showed a persistent, but greatly reduced, seasonal signal in some time series (*SI Appendix, Table S2*).

EWI Analysis. In 14 of 16 analyzed time series, loss of resilience before the breakpoint was signaled by at least one of the four EWI metrics (coded “+” in Table 2; details provided in *SI Appendix, section S3*), based on the median of the trend distribution across yearly increments of rolling window sizes (*Methods*). We found trends towards increasing AR-1 and DR in 10 cases each (63%), toward increasing SD in nine cases (56%), and toward increasing or decreasing SK (according to the direction of the state change) in seven cases (44%) (Fig. 2 and Table 2). In some cases, AR-1 trends were increasing from negative values to positive values (*SI Appendix, section S3*). We recorded these trends as positive [coded “(+)” in Table 2]. In several cases, indicators showed trends opposite to the theoretical expectation (coded “−” in Table 2). Only the state variable phytoplankton biomass in LMS showed the theoretically expected trends in all EWIs, whereas in two state variables—nonalgal attenuation in LVM and cyanobacteria biomass in LW—all EWIs failed. The agreement was low between positive AR-1 and SD trends (five cases), but higher between positive AR-1 and DR trends (10 cases). Logistic regression revealed no significant relationships ($P < 0.05$) between EWI behavior and mechanism (competition, IGP, or trophic cascade), state variable level (species, group, or ecosystem), step change height, length of prebreakpoint time series, nor sampling interval (fortnightly or monthly).

Robustness to Rolling Window Size and Significance of EWI Trends.

We estimated the robustness of EWI trends to the size of the rolling window, shown as the distribution of trends around their median (boxplots in Fig. 2 for AR-1 and SD and in *SI Appendix, Fig. S4* for DR and SK). The large majority of trends were qualitatively robust to rolling window size, as demonstrated by the low number of trend distributions in which the boxplot crossed zero on the y axis.

We also tested for the significance of the trends by estimating the rate of false positives using simulated surrogate time series (*Methods*). Among all 64 EWI trends (16 time series by four EWIs), only three trends were significant in >50% of the comparisons between data-based and surrogate-based trends. An additional 29 EWIs were significant in <50% of comparisons, and 32 EWIs showed no significant differences between data-based and surrogate-based trends (gray bars in Fig. 2 and *SI Appendix, Fig. S4*).

Table 1. Summary of case study systems, state variables, and drivers by mechanism: Trophic cascade, intraguild predation, and competition

Mechanism	Case study	Shift in state variable	Driver	Process	References
Trophic cascade	LW	Increase in water transparency	Increase in grazing pressure	Trophic cascade from longfin smelt (<i>Spirinchus thaleichthys</i>) on secondary consumer <i>Neomysis</i> on primary consumer <i>Daphnia</i> and finally on producers (phytoplankton)	(53, 54)
	LMS	Increase in <i>Dreissena polymorpha</i> larvae	Decrease in predation pressure	Changes in dominant carnivorous zooplankton species coincided with increase in <i>Dreissena</i> larvae.	(55, 56)
		Decrease in <i>Leptodora kindtii</i> (period 1)	Increase in predation pressure	<i>Leptodora</i> are a preferred prey for fish and likely indicate changes in overall fish predation pressure	
		Increase in <i>Leptodora kindtii</i> (period 2)	Decrease in predation pressure		
Competition	LMS	Decrease in phytoplankton biomass	Reoligotrophication	Reduction in nutrients decreases phytoplankton growth, improving light climate favoring macrophyte reestablishment	(57, 58)
		Decrease in <i>Aphanizomenon</i> (period 1)	Reoligotrophication counteracted by spring warming	Warmer springs promote cold-adapted cyanobacteria development	(59)
		Decrease in <i>Aphanizomenon</i> (period 2)	Reoligotrophication	Reduction in nutrients decreases phytoplankton growth, improving light climate favoring macrophyte reestablishment	(57)
	LW	Decrease in non- <i>Daphnia</i> cladocera	Reduction in predation pressure on <i>Daphnia</i>	Indirect effect of trophic cascade through increasing resource competition by <i>Daphnia</i>	(53, 54)
		Increase in cryptophytes	Reoligotrophication	Reduction in nutrients decreases cyanobacteria competitive ability and releases other phytoplankton from competition	
		Decrease in cyanophytes	Reoligotrophication		
	LZL	Decrease in water transparency	Eutrophication, epiphyte shading, and herbivory reduction in nutrients and benthivorous fish	Competition between submerged vegetation and phytoplankton under eutrophication and herbivory on macrophytes	(60, 61)
	LVM	Decrease in nonalgal attenuation		Recovery of submerged vegetation cover and subsequent stabilization of sediments	(62)
	LV	Increase in functional group P, decrease in functional group U	Eutrophication	Competition between functional groups P (eutrophic epilimnion species) and U (summer epilimnion species)	(63, 64)
		Increase in functional group H1	Reoligotrophication	Competition between nonnitrogen-fixing and dinitrogen-fixing species (functional group H1)	
Intraguild predation	LMS	<i>C. vicinus</i> – <i>C. kolensis</i> dominance switch	Reduction of shared food source	The inferior resource competitor <i>C. vicinus</i> preys on juveniles of the smaller <i>C. kolensis</i>	(32)

Case study systems are Lake Müggelsee (LMS), Lake Veluwemeer (LVM), Lake Zwemlust (LZL), Lake Washington (LW), and Lake Vörtsjärv (LV).

EWIs in Stepwise Shortened Time Series. To assess whether EWI trends could have been detected in incomplete time series that would have been available at 1 y or longer before the actual transition took place, we repeated the EWI analysis on stepwise shortened (i.e., yearly increments) time series of each state variable. In 14 of the 16 time series, at least one of the EWIs would have indicated a loss in ecosystem resilience between 1 and >10 y before the critical transition. In three instances, positive EWIs trends were seen in only the year before the shift (*SI Appendix, Table S5*), whereas in 17 instances, the median of the EWI trend distribution remained positive (i.e., showing the expected sign) up to the minimum time series period analyzed (3 y).

Discussion

In this study, we assessed the detectability of four commonly used EWIs (AR-1, SD, SK, and DR) in empirical time series of freshwater ecosystems. By combining high-quality empirical time series with ecological understanding and standardized methods, we have shown that EWIs preceded critical transitions in natural aquatic ecosystems, in some cases even several years ahead of the shift, despite the potential shortcomings of empirical datasets, such as observation error, sparse sampling, and low signal-to-noise ratio (9, 24, 33). However, in a large proportion of cases the EWIs failed, and the cases with positive EWIs generally showed low or no significance. In addition, we found little agreement among signals,

Table 2. Overview of all 14 state variables (16 time series) listing the direction of the shift, timing of the breakpoint (year-month), step height (scaled to SD), two-sample Welch test significance, and for each EWI the sign of the median of the trend distribution (across all rolling window sizes)

Case study	State variable	Direction of shift	Breakpoint (y-mo)	Step (scaled to SD)	t (Welch)	AR-1	SD	SK	DR
LMS	Phytoplankton, mg L ⁻¹	Decrease	1990-5	0.803	9.92	+	+	+	+
	<i>Aphanizomenon</i> , mg L ⁻¹ P1	Decrease	1990-11	0.413	4.47	+	+	-	+
	<i>Aphanizomenon</i> , mg L ⁻¹ P2	Decrease	2002-12	0.317	5.92	-	+	+	-
	<i>Cyclops vicinus</i> , ind L ⁻¹	Decrease	1992-6	0.629	7.86	+	-	+	+
	<i>Dreissena polymorpha</i> larvae, ind L ⁻¹	Increase	1993-9	0.388	-6.79	+	+	-	+
	<i>Leptodora kindtii</i> , ind L ⁻¹ P1	Decrease	1987-11	0.776	6.58	-	+	-	-
	<i>Leptodora kindtii</i> , ind L ⁻¹ P2	Increase	2005-7	0.239	-5.21	(+)	-	-	+
LW	Secchi depth, m	Increase	1976-10	1.29	-21.94	+	+	-	+
	Cryptophytes, 100 μm ³ L ⁻¹	Increase	1977-7	0.627	-13.16	(+)	-	-	+
	Cyanobacteria, 100 μm ³ L ⁻¹	Decrease	1973-4	1.587	9.33	-	-	-	-
	Non- <i>Daphnia</i> cladocerans, ind L ⁻¹	Decrease	1976-8	0.783	6.09	-	+	+	-
LV	H1, mg L ⁻¹	Increase	1992-8	0.569	-6.75	(+)	+	-	+
	P, mg L ⁻¹	Increase	1977-12	0.281	-4.61	(+)	-	+	+
	U, mg L ⁻¹	Decrease	1977-12	0.532	3.17	(+)	-	+	+
LVM	Nonalgal attenuation, m ⁻¹	Decrease	1995-9	1.058	9.05	-	-	-	-
LZL	Secchi depth, m	Decrease	1992-9	1.098	7.97	(-)	+	+	-
Proportion of EWI trend distribution medians corresponding to the theoretical trend expectation						10/16	9/16	7/16	10/16

The t (Welch) values in bold type are significant at the $\alpha = 0.05$ level. The tested EWIs are autocorrelation at lag-1 (AR-1), standard deviation (SD), skewness (SK), and density ratio (DR). Median trends that corresponded with the theoretical expectation are coded with "+"; trends that showed the opposite of expected with "-"; brackets denote AR-1 trends crossing 0 on the scale.

and observed no relationship between EWI trends and potential predictors, such as the ecological mechanism of the shift, state variable level, magnitude of the transition, or sampling interval.

Use of Ecological Understanding in the Choice of State Variables. The choice of case studies and state variables in this study was based on expert knowledge of ecosystems that likely experienced critical transitions. Although this choice was based on an understanding of the ecological mechanisms that can give rise to alternative states in aquatic ecosystems, it does not provide conclusive evidence that the regime shifts that we analyzed correspond to true critical transitions. Nonetheless, such a priori choices of state variables for EWI analysis implicitly excluded assumptions based on other transition types, such as responses to step changes in the driver (34), but offered clear expectations of the EWI behavior that should precede a transition (25). Because mechanisms generating critical transitions can operate at all ecosystem levels, relevant state variables range from discrete variables, such as species-specific biomass (e.g., ref. 32), to aggregated variables, such as Secchi depth, turbidity, or metabolism (e.g., ref. 12). In addition, the case studies were chosen based on the availability of long time series of monitoring data for relevant state variables at sufficiently high temporal resolution and with few missing values (35). Despite our conscientious choice of ecosystems and state variables, we still found contradicting patterns in EWIs.

Detection and Agreement Among Indicators. Not all EWIs were equally reliable in detecting impending transitions (Table 2). In 44% of the 64 cases, the EWI failed. Only in three cases where signals were positive were trends significant in >50% of data-based and surrogate-based trend comparisons. Our significance testing relied on estimating EWI trends in surrogate stationary data fitted to the original time series to determine the rate of false positives. Unfortunately, we lacked records to serve as controls when comparing trends from comparable aquatic systems in which no transition occurred to estimate the rate of true negatives (no alarm, or sensitivity) of the EWIs. Given the lack of such controls, one potential way to measure the no alarm rate is to derive trends from nonstationary models fitted to the data

(36). In future work, it would be valuable to compare trends in EWI in study systems that did not exhibit any changes or in which changes were brought about by large external shifts in drivers.

Agreement between AR-1 and SD trends has been postulated as a minimum requirement to signal the approach of a transition (8). In our dataset, these two indicators concurrently increased in only 5 of 16 time series. Critical transitions with increasing AR-1 and decreasing SD have been observed in other studies as well (36). Such inconsistent trends between indicators may occur in ecosystems that are subject to multiple concurrent regime shift-generating processes that may or may not interact via shared state variables and that may react differently to drivers and environmental noise (9). If such connected regime shift processes work toward muffling variance in the measured state variable, then the variance-based EWI signal could be suppressed (37). Transitions from cycles to stable points also can generate decreasing variation (38) and may explain decreasing SD trends like in the LMS *Cyclops vicinus* abundances or the LV functional group U biomass.

In many cases, autocorrelation (AR-1) was generally low, and in some cases it was even negative (e.g., LW non-*Daphnia* cladocerans). Low AR-1 values (far from the theoretical value of 1 at which critical transitions occur) reflect the fact that transitions in the real world are likely triggered well before the actual tipping point is reached (25, 39). Evaluation of the full power spectrum for changes in power in aggregated low vs. high frequencies over time indicated that higher-order AR processes did not provide more information than the AR-1. This finding suggests that the reported low or negative autocorrelation in our records likely originated from too long sampling intervals in cyclic variables (e.g., population cycles), resulting in undersampled cycles of fast-growing plankton and its related variables.

Successful detection of EWIs often has been related to the availability of high sampling frequency data (40), although it has been shown that EWIs still can be detected robustly in infrequently sampled data as long as the time series are sufficiently long (41). Our study was based on data sampled at (or averaged to) fortnightly and monthly intervals. Monthly intervals are rather long compared with the generation or reaction times of the state

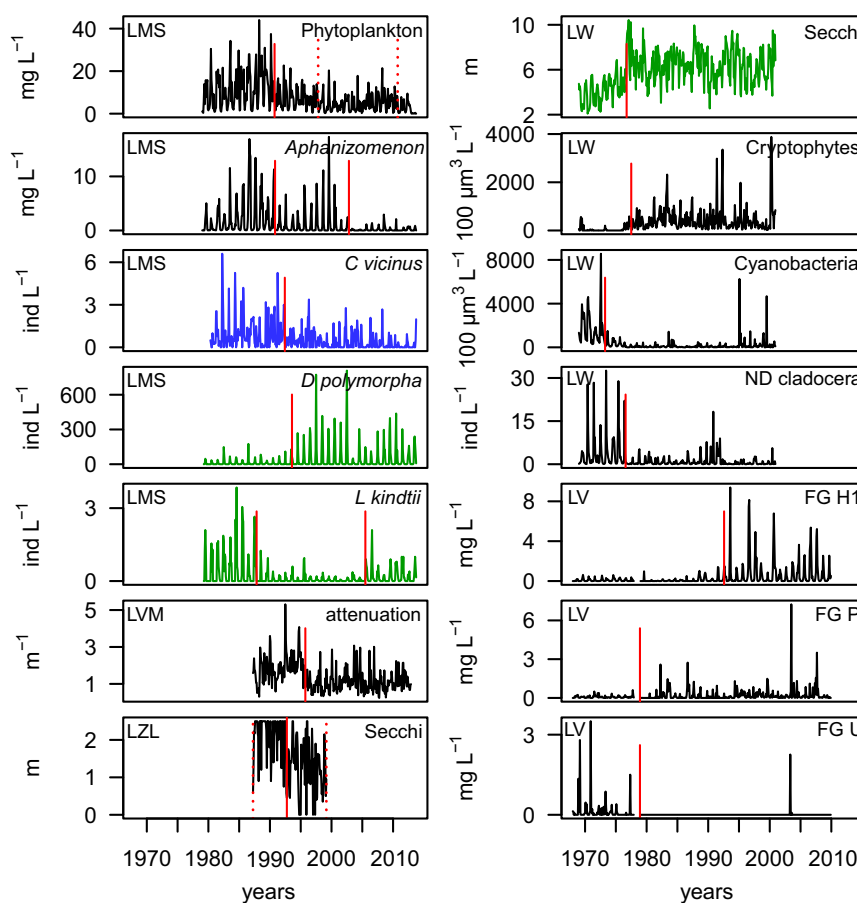


Fig. 1. Time series of 14 selected state variables (black lines, competition; green lines, trophic cascade; blue line, intraguild predation) of five lakes. The solid red lines indicate the timing of critical transitions (estimated by breakpoint analysis; *Methods*). The dashed red lines indicate additional data-based breakpoints that were not used in this study because they either were not described in the literature (LMS phytoplankton biomass) or resulted from biomanipulation (LZL Secchi depth).

variables tested in our study; for example, phytoplankton generation times are on the order of days, and thus monthly averages of biomass aggregate multiple generations of phytoplankton. Nevertheless, we could not detect a relationship between sampling interval and the proportion of failing EWs in our dataset.

Finally, data preparation, particularly choices regarding detrending and seasonal adjustment methods, can affect the autocorrelation structure in time series and hence the outcome of EWI analyses. We used Gaussian filtering with a fixed bandwidth based on the sampling frequency to detrend and seasonally adjust the data before EWI analysis (*SI Appendix, section S6*); however, despite this data preparation, some time series still showed reduced traces of seasonality. Remaining seasonal signals may increase or decrease the intercept of the EWI trends, but not the sign of the trend.

Early Detection Based on Incomplete Time Series. Informative changes in some of the EWI metrics were already detectable several years before the actual transition, although large differences in detection windows between state variables were observed. These differences may be explained in part by the fact that the actual shift is usually triggered by external stochastic perturbations, which in turn are often independent of the drivers of ecosystem stability loss (42). However, in many cases EWIs indicated sustained instability over the period tested in our study, which may be attributed to either an insufficient time span available for testing (e.g., LZ) or unrecognized interacting processes that promoted prolonged instability in these ecosystems.

Conclusion

Despite our informed and conscientious choice of case study ecosystems and state variables, we found relatively low detectability of EWIs before the documented transitions, and when EWIs were detected, the agreement among them was low. Our findings are in line with results from an assessment of the detectability of EWIs before nonlinear transitions (43). Although it is encouraging that we could detect EWIs in some of our empirical aquatic time series using data derived from commonly used monitoring schemes that were not designed for this purpose, the lack of reliability and agreement among signals limits the potential application of EWIs only to well-understood ecosystems (35). In such well-understood ecosystems, harnessing EWIs as metrics of loss of resilience may aid planning for the unpredictable and could be part of strategic foresight programs for management and conservation (44); however, our analysis suggests that these metrics can be of added value only in combination with existing frameworks (e.g., alternative stable-state theory) and in-depth ecosystem knowledge. Furthermore, taking into account the underlying assumptions and requirements of EWI analysis can inform managers about adaptations in monitoring schemes by advising about relevant variables and temporal sampling resolution to adequately capture changes in the resilience of systems.

One way forward may be the advent of automated, continuous high-frequency monitoring, ideally monitoring multiple lakes with similar properties for comparison (35). To increase our

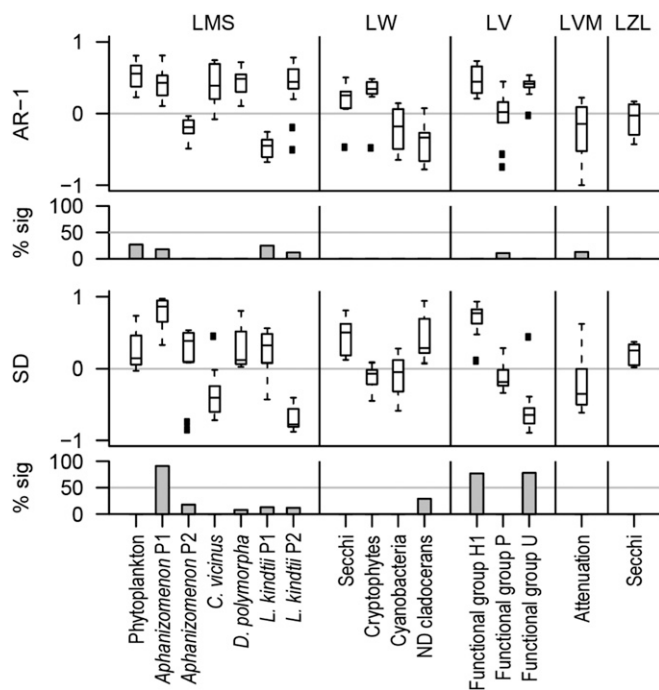


Fig. 2. Robustness and significance testing. Mann-Kendall trend distributions for EWIs AR-1 and SD (panel) and per analyzed time series (boxplots) and the proportion of significant differences in trends between data-based and surrogate time series across all rolling window sizes (significance, gray bar plots). Positive values of trends in the boxplots represent increasing EWIs trends. Note that in almost all time series, the trends were robust to the choice of rolling window size. Significance testing was based on comparing indicator trends produced from stationary surrogate time series to the empirically reported for fixed rolling window size (Methods).

understanding of critical transition generating processes, ecosystem models, such as PCLake, can help bridge the gap between simple minimal models and the full complexity of natural systems and allow combined analysis of, e.g., food-web theory and alternative stable states theory and indicators of ecosystem resilience (45). Ideally, such insights can serve to broaden our search image in empirical EWIs patterns instead of relying on a generic increase in variance without understanding the inherent variability in ecosystems. In the meantime, the reliability of EWIs for predicting abrupt shifts in ecosystem state should be viewed with caution.

Methods

Data Preparation and Breakpoint Detection. The data preparation and statistical analysis protocol was identical for all 16 state variable time series. Each time series was analyzed at the highest temporal resolution available or at a lower resolution that resulted in less missing data (SI Appendix, Table S2). LMS and LW were analyzed at fortnightly; all other case studies were analyzed at monthly intervals. Because our time series methodology requires continuous and equidistant data, we imputed missing values up to a maximum of four consecutive time steps using a Kalman filter (46). Time series with longer gaps were shortened to start or end at the gap (e.g., LV). Each time series was standardized by mean centering and SD scaling for convenient comparison of step change heights. The timing of the step was determined by breakpoint analysis, given that step changes in the respective state variables may differ from the timing of whole-system step changes reported in the literature. Robust estimates of the timing of step change were achieved using three complementary breakpoint estimation methods: (i) additive decomposition of time series in seasonal, trend, and residual components and subsequent iterative fitting of piecewise linear season and trend models (47) using the R package “bfast” (48); (ii) testing for step changes in the average using the Pettitt test (49); and (iii) STARS, a combination of a sequential partial CUSUM method and a t test (50). If at least two methods showed similar timing for a step change (± 12 mo), then the stan-

dardized original time series was split at that breakpoint. Differences between prebreakpoint and postbreakpoint means of the time series were tested with the Welch two-sample t test (51) (Table 2).

Because of a 1-y gap in the time series, breakpoint timing of two state variables—LV functional groups P (eutrophic epilimnion species) and U (summer epilimnion species)—was determined differently. Here we assessed whether large changes in the biomasses of the functional groups occurred during the gap, indicating a potential shift. The pre-1978 and post-1978 time series showed significantly different means for functional group U (Welch two-sample t test, $t = 3.52$; $df = 128$; $P < 0.001$; difference in means, $0.29 \times SD$) and functional group P (Welch two-sample t test, $t = -4.67$; $df = 439$; $P < 0.001$; difference in means, $0.55 \times SD$), suggesting that the shift in functional groups U and P occurred during the year 1978 (Table 2). Because both of these time series showed no further breakpoints in the years after 1978, we conservatively assumed 1977 to be the transition year.

EWI Analysis. The prebreakpoint time series was seasonally adjusted using a Gaussian smoother with a kernel bandwidth based on the number of data points per year (i.e., 12 for monthly datasets and 26 for fortnightly datasets; other methods are compared in SI Appendix, section S6). A bandwidth of 1 y was chosen to account for annually recurring patterns while retaining informative low- and high-frequency variability other than long-term trend and season (15). In three residual time series—LMS *Dreissena* (two outliers) and *Aphanizomenon* period 1 and LW Cyanobacteria (one outlier each)—local outliers were replaced by Kalman-imputed values. The residuals were then passed on to analysis of EWIs AR-1, SD, SK, and DR with testing for robustness to the size of rolling window and testing for significance (false positives) using the R package “earlywarnings” (9). In addition, the static choice of fixed compared frequencies in the EWI DR was confirmed to be sufficiently capturing changes of the full power spectrum of the prebreakpoint residuals time (based on its estimated smoothed fast Fourier periodogram).

Robustness and Significance Testing. The trends in the estimated temporal evolution of EWIs from the rolling window approach were quantified using the nonparametric Mann-Kendall trend test, which tests for monotonic trends based on the Kendall τ rank correlation coefficient (9). Because the size of the rolling window can affect EWI trends (36), a robustness analysis was performed to estimate the distribution of trends and proportion of trends that did not differ in sign from the median of the trend distribution. This was done using yearly increments of the residuals time series covered by the rolling time window [2 to $n - 2$ y, function “sensitivity_ews” (9), where n is the number of years in each prebreakpoint time series]. Significance testing was done by comparing the data-based EWI trend against a bootstrapped distribution of 200 surrogate time series-based EWI trends. The surrogate time series were generated based on an ARMA(p,q) model fitted on the residuals time series. The bootstrapped distribution of trends depicts the probability that a particular trend could occur by chance in time series of the same ARMA structure [function “surrogates_ews” in R package “earlywarnings” (9)]. The data-based EWI trend was deemed significant if it fell on one of the 5% tails of the surrogate-based trend distribution ($\alpha = 0.1$). This significance testing was repeated for all rolling window sizes, and the proportion (%) of significant trends over all rolling window sizes was reported.

Relationship of EWIs with State Variable Categories and Agreement Among Signals. The relationships between the occurrence of each EWI and the predictor variables mechanism (competition, IGP, or trophic cascade), state variable category (species, group, ecosystem), step change height, length of prebreakpoint time series, and sampling interval (fortnightly or monthly) were analyzed by logistic regression [identity link for continuous predictors (step change height and time series length); logit link for categorical predictors] with Bonferroni-corrected post hoc testing. The median of the trend distribution across all time window sizes was used to assess agreement among EWIs.

Early Detection of EWIs. To assess the potential timeliness of EWIs in detecting the approaching transition, EWI trends were quantified on stepwise shortened prebreakpoint time series, starting with the full time period and continuing with stepwise reduction of the time series by cutting off the last year of data. The minimum time series length was set to 3 y to allow for meaningful Kendall τ estimation. The number of years before the shift that an EWI could have been detected was set by assessing the maximum number of stepwise reductions before the theoretically expected EWI trend disappeared (i.e., when the median of the Kendall τ trend distribution turned negative for AR-1, SD,

and DR or changed sign for SK). All data analyses and graphing were conducted using the R language environment for statistical computing (52) and associated library extensions.

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Supplementary information

The supplementary information to “Evaluating early-warning indicators of critical transitions in natural aquatic ecosystems” contains six sections.

S1: Case study descriptions provides more information on each of the five lakes that serve as case studies in this paper.

S2: Table S2 provides an overview on the data structure of the time series that serve as state variables in the paper, including information on the overall period covered, sampling / aggregation interval, number of pre-breakpoint time series data points, percentage of missing data and remaining seasonal ARIMA structure after seasonal adjustment by Gaussian smoother.

S3: Overview figures provides three figures per state variable time series. The first figure shows the evolution of EWI trends (window size 50%) and sensitivity to window size (i.e. robustness) as well as the proportion of significant differences in trends between data-based and surrogate time series. The second figure shows how early EWI trends would have indicated a transition using subsequently shortened time series. Here we show boxplots of EWI trends (varying window sizes) and associated sensitivity to window size (i.e. robustness) and proportion of significant differences in trends between data-based and surrogate time series. The third figure shows the ACF, PACF and periodogram of the original and the seasonally adjusted pre-breakpoint time series to allow assessment of the effect of seasonal adjustment on the remaining seasonal signal in the time series.

S4: Figure of robustness and significance in EWIs skewness (SK) and density ratio (DR) shows the trend distribution (boxplots), the robustness to varying window size (spread of boxplots) and the proportion of significant differences in trends between data-based and surrogate time series (barplots) for each state variable.

S5: Table S5 Summary of early detection of EWIs provides an overview of how many years before the transition an EWI would have been detected.

S6: Comparison of detrending and seasonal adjustment methods shows a comparison of six different methods using modelled data of the ecosystem model PCLake.

S1: Case study descriptions

Lake Müggelsee (LMS)

Re-oligotrophication driven change in competition, food availability driven change in intraguild predation and trophic cascade driven change in predation pressure

The Müggelsee dataset spans a total of 34 years, starting in January 1979. Müggelsee is a shallow (mean depth 4.9 m, max depth 8m), polymictic and eutrophic lake situated southeast of Berlin (Germany) with a surface area of ~750 ha and a retention time of about 6-8 weeks (1, 2). A detailed description of the regular sampling procedure and sample processing is provided in Gerten and Adrian (3). Köhler *et al.* (1) have described three distinct eutrophication periods in the lake: a hyper-eutrophic period (1979-1990) characterised by turbidity and loss of submerged vegetation; a transient period (1991-1996) characterised by internal loading of phosphorus, a reduction in phytoplankton biomass, reappearance of spring clear-water phases, and slow recovery of submerged vegetation (2); and a eutrophic period (1997-2003) characterised by slowly decreasing total nitrogen (TN). However, the lake has not fully shifted to a clear state, probably due to interfering processes such as a) the reduction in TN leading to a competitive advantage for N-fixing cyanobacteria in summer conditions; b) climate warming leading to increased internal loading as well as to an increase in stratification events that promote cyanobacterial blooms; c) constant grazing pressure by waterfowl and fish inhibiting recovery of macrophytes; or d) potentially increasing cyprinid fish population due to reduced commercial interest after the German reunification, leading to increased predation pressure on zooplankton in the period 1990 to probably 2005 (1, 2, 4).

Three mechanisms leading to regime shifts at different ecosystem levels appear important in Müggelsee, the eutrophication / re-oligotrophication driven reduction in phytoplankton biomass; a food-availability driven shift in an intraguild predation system of two copepod species; and a predation pressure driven shift in *Dreissena polymorpha* larvae and in *Leptodora kindtii* abundances. Five variables were selected for early-warning signal (EWI) analysis, two pertaining to the re-oligotrophication driven shift in phytoplankton biomass: total phytoplankton biomass and *Aphanizomenon* biomass (1, 5); one time series pertaining to the food-availability driven shift in the *Cyclops vicinus* and *Cyclops kolensis* intra-guild predation system (6) and two time series of shifts in *Leptodora kindtii* and in *Dreissena polymorpha* larvae which may have been caused by changes in fish predation pressure on carnivorous zooplankton cascading on to *Dreissena* larvae (4, 7) although their rise has also been linked to earlier spawning under warming (8, 9). Testing the *Cyclops kolensis* abundance time series was not possible as the pre-breakpoint time series was severely censored by a detection limit. Overall the dataset had few missing values except for the first period of *L. kindtii*. We chose to include this time series nonetheless as the missing values occurred during winter periods, a period in which *L. kindtii* are known to not occur in the pelagic. Two

outliers in the *Dreissena* residuals time series and one outlier in the *Aphanizomenon* residuals time series (Period 1) were removed and replaced with Kalman imputed values. For a summary on number of data points and percentage of missing data see Table S2.

Lake Zwemlust (LZL)

Eutrophication driven change in competition

The full Zwemlust dataset encompasses 16 years, starting in May 1986 and ending in August 2001.

Zwemlust is a small (~1.5 ha), shallow (mean depth 1.5m), eutrophic and polymictic lake situated close to the city of Utrecht (The Netherlands). Zwemlust has no surface in- or outflow but receives high external phosphorus (P) and nitrogen (N) loadings by seepage from the nearby River Vecht (10) leading to persistent phytoplankton blooms and low water transparency prior to 1987. The lake was restored twice by biomanipulation (March 1987 and April 1999). These two dates also represent the boundaries of the data analysed here, see also Table S2. The first biomanipulation in 1987 included draining the lake, removal of planktivorous fish, stocking of piscivorous fish, creation of zooplankton shelter and planting of macrophytes. These manipulations were successful in shifting the lake into a macrophyte dominated, clear state. However, Zwemlust experienced a return of low water transparency periods in summer and a loss of submerged vegetation in 1992. In 1999, the second, less severe biomanipulation successfully shifted the system back to a clear state (10). The loss of system resilience was explained by a combination of selective herbivory by coots (*Fulica atra*) promoting a macrophyte succession to species less resistant to epiphyte shading (10) in combination with epiphyte growth due to continuing high phosphorus availability (11). Secchi depth was selected to test this eutrophication driven regime shift for EWIs.

Lake Veluwemeer (LVM)

Eutrophication and light climate driven change in competition

The Veluwemeer dataset spans almost 26 years from April 1987 to December 2012 (12). Veluwemeer is a shallow (mean depth of 1.5 m), polymictic lake with a surface area of 3 05 ha in central Netherlands. The lake is directly connected to Lakes Wolderwijd and Drontermeer. Ibelings *et al.* (13) identified four distinct phases: i) Initial clear state (1957-1970) characterised by increasing eutrophication by waste water effluent and run off from adjacent agricultural lands, widespread loss of macrophyte cover and a fish community change to bream dominance. Increased turbidity was caused by cyanobacteria blooms (mainly *Planktothrix agardhii*) and wave re-suspension of particles. ii) Turbid state (1971-1985) characterised by reduction of external P loading by implementing waste water treatment plants and winter flushing of the lake with calcium rich polder water, resulting in a reduction in chlorophyll but not in turbidity due to resuspension of sediment particles. iii) Transient period (1986-1995) characterised by the co-existence of

clear (above *Chara* stands) and turbid (above deeper lake areas) patches after a series of severe winters helped break the dominance of *Planktothrix* sp. blooms. From 1994 onwards, *Chara* stands expanded rapidly in parallel to a reduction in the benthivorous fish population and the return of *Dreissena*. iv) Clear state (1996-2012) stabilised by a high *Chara* coverage, high densities of *Dreissena* and a fish community composition change to planktivorous species dominance resulting in a stabilisation of the sediment and a reduction in non-algal light attenuation (i.e. the non-chlorophyll related turbidity, see (13)). The clear state seems to be resilient as it has recovered from perturbations such as dredging and record precipitation. Non-algal light attenuation was selected to test this eutrophication driven regime shift for EWIs.

Lake Washington (LW)

Re-oligotrophication driven change in competition and release from overexploitation trap

The dataset for Lake Washington analysed here spans 27 years from January 1969 to December 2000. Lake Washington is a large (~8 800 ha) and deep (maximum depth ~65 m) lake within the city of Seattle, WA, see Table S2. Lake Washington experienced modest eutrophication from the 1940s to 1968 by sewage water inflow. Secchi depth was reduced, and *Oscillatoria* blooms were common (14). Since 1968 wastewater is diverted from the lake which greatly reduced the input of external P to the lake and had an immediate effect on the phytoplankton biomass production and Secchi depth. By 1972 the lake was considered recovered (15). In 1976, Secchi depth improved further, which was attributed to the establishment of *Daphnia* and increased grazing pressure on the summer phytoplankton bloom (16). The sudden establishment of *Daphnia* in the lake was attributed to a trophic cascade releasing *Daphnia* from an overexploitation trap and a P-reduction related decrease in low food quality *Oscillatoria* blooms. The trophic cascade was probably initiated by an unintentional improvement of spawning grounds for longfin smelt (*Spirinchus thaleichthys*) during flood-control measures on the lake's main tributary (17). These fish started preying heavily on *Neomysis*, releasing *Daphnia* from predation pressure (15, 18). The increase of more palatable phytoplankton groups such as cryptophytes co-occurred with the disappearance of the poor food quality species *Oscillatoria* (19) from the lake in 1976 (15), which may have resulted from changed competitive ability of these two groups under nutrient ratio changes in the lake in the wake of the waste water diversion since 1968.

The period of strong nutrient reduction due to improved sewage management from 1963 -1968 was excluded from the analysis as a biotic response to such a strong external change would not fall into the category of critical transitions (i.e., it is a large response primarily to large changes in an external forcing variable). Hence the analysis focussed on the biotic response of the cladoceran competitors *Daphnia* and non-*Daphnia* cladocerans to the trophic cascade and on the response of the phytoplankton competitors

Cryptophytae and *Oscillatoria* in the period after sewage diversion. The assumed loss of *Neomysis* due to the trophic cascade could not be observed as data on night-time pelagic *Neomysis* abundances in the lake were not available. Moreover, the time series of *Daphnia* was not analysed as pre-shift *Daphnia* abundances were mostly below detection limit. One outlier was removed and replaced by a Kalman imputed value in the residuals time series of cyanobacteria.

Lake Võrtsjärv (LV)

Competition under environmental change

Lake Võrtsjärv is a large (~27 000 ha), shallow (mean depth ~2.8 m), eutrophic and polymictic lake situated in central Estonia. The lake is connected to six inflows and one outflow to Lake Peipsi. The lake experienced an increase in water levels from 1964 up to the early 1990s, and a decrease in water levels after the early 1990s; however, large fluctuations in water levels have occurred throughout. High water levels in combination with increased colouring of the water by humic matter decreased the area available for the colonisation by submerged macrophytes. The reconstructed eutrophication history of the lake showed an increase in loading since the 1950s with peak loadings during the 1980s (20) exposing the lake to increasing eutrophication. Phytoplankton biomass increased with eutrophication and peaked in the 1970s (21). In 1977/78 a sudden increase in water level brought about a complete and permanent reconfiguration of the phytoplankton community composition and a marked decrease in summer biomasses. Summer epilimnion species (functional group U) such as *Uroglena* sp (22) declined suddenly and eutrophic epilimnion species (functional group P) such as *Aulacoseira granulata* increased abruptly. Moreover, the turbid mixed layer species (functional group S1) changed from a *Planktolyngbya limnetica* and *Aulacoseira ambigua* dominated group to a dominance of more shade resistant but also shade generating phytoplankton species such as *Limnothrix* sp. (21). A sudden increase in 1993 of di-nitrogen fixing species such as *Aphanizomenon skujae* (functional group H1) was attributed to declining N:P ratios particularly at low water levels which promoted denitrification and P release from sediment (21).

To reflect the functional changes in the plankton community, functional (Reynold) groups were selected for further analysis, in particular the groups U (summer epilimnion species such as *Uroglena*, which tolerate low nutrients but are sensitive to CO₂ limitation), P (eutrophic epilimnion species such as *Fragilaria crotonensis* or *Aulacoseira granulata*, which tolerate mild light and C limitation but are sensitive to stratification and Si deficiency) and H1 (dinitrogen-fixing Nostocaceae such as *Aphanizomenon* or *Anabaena flos-aquae*, which tolerate low nitrogen and carbon but are sensitive to poor light, low phosphorus and mixing) (22).

S2: Table S2

Case Study System	State Variable	Overall Time Period	Interval	# Data Points	% Missing Data	Comments on missing data	Seasonal ARIMA AR, I, MA
LMS	Phytoplankton	1979 - 1990	bi-weekly	295	2.4		1,0,0
	<i>Aphanizomenon</i> P1	1979 - 1990	bi-weekly	308	2.7		0,0,2
	<i>Aphanizomenon</i> P2	1990 - 2002	bi-weekly	313	2.9		1,0,2
	<i>Cyclops vicinus</i>	1980 - 1992	bi-weekly	315	16.2	Winter bi-weeks 1-6	0,0,0
	<i>Dreissena polymorpha</i> larvae	1979 - 1993	bi-weekly	374	15.5	Winter bi-weeks 1-6	0,0,0
	<i>Leptodora kindtii</i> P1	1979 - 1987	bi-weekly	222	24.4	Winter bi-weeks 1-6	1,0,1
	<i>Leptodora kindtii</i> P2	1987 - 2005	bi-weekly	461	3.3		2,0,0
LW	Secchi depth	1969 - 1976	bi-weekly	202	1.5		0,0,2
	Cryptophyceae	1969 - 1977	bi-weekly	222	4.9		1,0,1
	Cyanophyceae	1969 - 1973	bi-weekly	110	9.1		1,0,1
	Non- <i>Daphnia</i> cladocerans	1969 - 1976	bi-weekly	198	6.1		1,0,1
LV	H1	1979 - 1992	monthly	163	11.6	Mostly winter months	1,0,0
	P	1968 - 1977	monthly	120	0.8		2,0,3
	U	1968 - 1977	monthly	120	0.8		1,0,1
LVM	Non-algal light attenuation	1987 - 1995	monthly	102	8.8		0,0,1
LZL	Secchi depth	1987 - 1992	monthly	66	4.5		0,0,0

Table S2: Summary of overall period covered, aggregation interval, number of pre-breakpoint time series data points, percentage of missing data with comments and remaining seasonal ARIMA structure after seasonal adjustment by Gaussian smoother

S3: Overview figures of early-warning indicators per state variable

Here we present three figures per time series showing early-warning indicators, sensitivity to window size (i.e. robustness); the proportion of significant differences between trends in data-based and surrogate time series for the full pre-breakpoint time series (upper figure) and across stepwise shortened pre-breakpoint time series (middle figure); and information about seasonal signals in the original and seasonally adjusted time series (lower figure)

Upper figure: Early-warning indicator trends, sensitivity and significance of the full pre-breakpoint time series

top panel: standardized pre-breakpoint time series (black) and Gaussian smoother (red)

second panel: residual time series (standardized pre-breakpoint time series - Gaussian smoother)

third panel: The evolution of the autoregressive coefficient of a first order AR model (AR-1) fitted on the data using a 50% rolling window. Theory expects that AR-1 increases in the approach to a breakpoint.

Bold font y-axis labels point out cases where AR-1 is partially negative. To the right are two box plots:

Right boxplot: Sensitivity (robustness): distribution of trends across all window sizes (yearly increments) of the original standardized pre-breakpoint time series “org”. Significance: distribution of trends based on 200 surrogate time series with the same ARMA structure “sg”. Mean differences and range of trend distributions show how much data-based trends differ from surrogate-based trends.

fourth panel: The evolution of standard deviation (SD) estimated on the data using a 50% rolling window. Theory expects that SD increases in the approach to a breakpoint. Boxplot as in AR-1.

fifth panel: The evolution of skewness (SK) estimated on the data using a 50% rolling window. Theory expects that SK shows a trend in the direction of the change at the breakpoint (i.e increase or decrease) in the approach to that breakpoint. Boxplot as in AR-1.

bottom panel: The evolution of density ratio (DR) of the power spectrum of the data estimated as the ratio of low frequencies over high frequencies using a 50% rolling window. Theory expects that DR increases in the approach to a breakpoint. Boxplot as in AR-1.

Middle figure: Early-warning indicator trends, sensitivity and significance of stepwise shortened time series

Top left panel: distribution of AR-1 trends (based on all window sizes) of the original standardized pre-breakpoint time series (grey) and of 200 surrogate time series with the same ARMA structure (white) across stepwise shortened pre-breakpoint time series (yearly increments, see x-axis). The proportion of trends with a slope significantly different from zero across all data-window sizes is shown in the bar-plot

“sens”, the proportion of significant differences ($\alpha=0.05$) between original and surrogate time series trends is shown in bar-plot “sig”.

Second left panel: distribution of SD trends (based on all window sizes) of the original standardized pre-breakpoint time series (grey) and of 200 surrogate time series with the same ARMA structure (white) across stepwise shortened pre-breakpoint time series (yearly increments). The associated bar-plots show sensitivity (i.e. robustness) to varying window size and significances.

Third left panel: distribution of SK trends across all window sizes of the original standardized pre-breakpoint time series (grey) and of 200 surrogate time series with the same ARMA structure (white) across stepwise shortened pre-breakpoint time series (yearly increments). The associated bar-plots show sensitivity (i.e. robustness) to varying window size and significances.

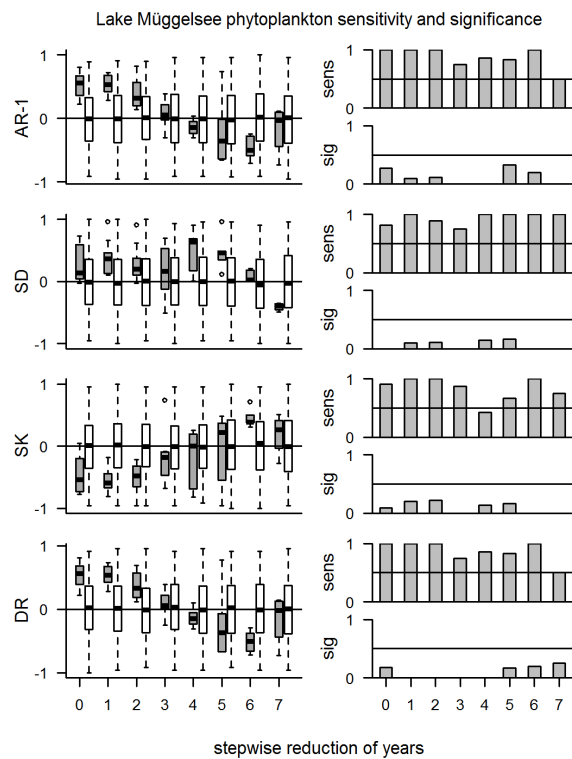
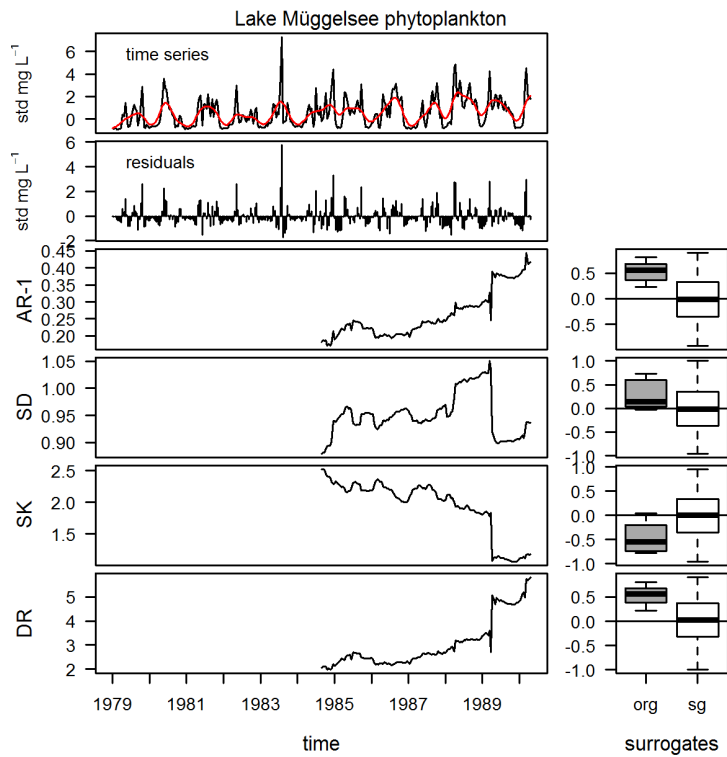
Bottom left panel: distribution of DR trends across all data-window sizes of the original standardized pre-breakpoint time series (grey) and of 200 surrogate time series with the same ARMA structure (white) across stepwise shortened pre-breakpoint time series (yearly increments). The associated bar-plots show sensitivity (i.e. robustness) to varying window size and significances.

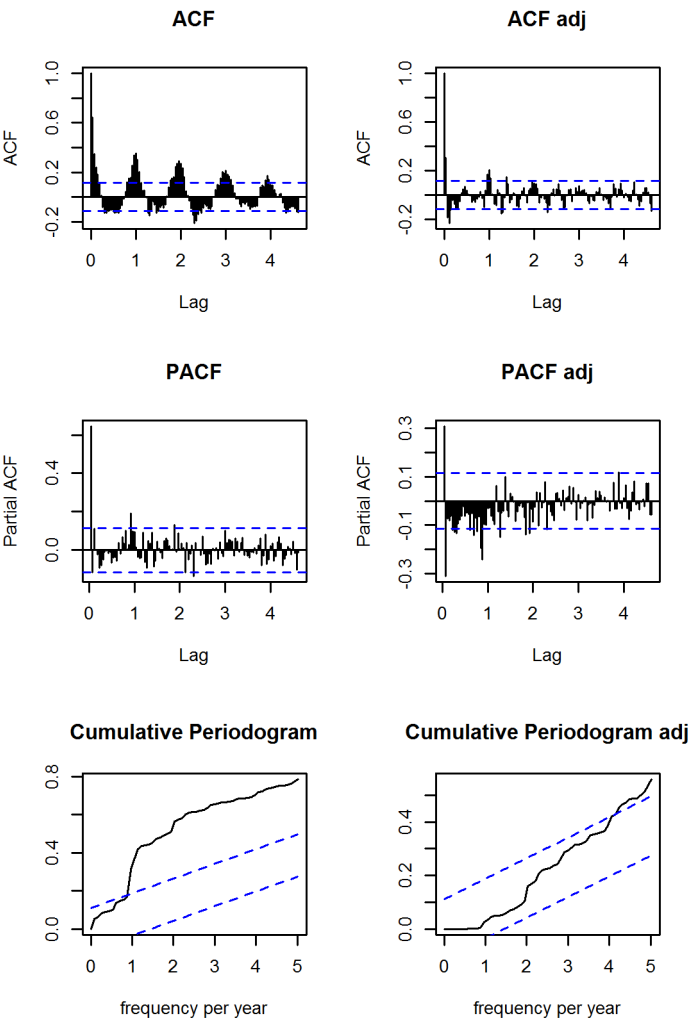
Lower figure: ACF, PACF and Periodogram of original and seasonally adjusted pre-breakpoint time series

Left panels from top to bottom: ACF (upper panel), PACF (middle panel) and the periodogram (lower panel) of the original (standardised and mean centered) time series up to the transition. Blue lines are 95% confidence intervals for a white noise process.

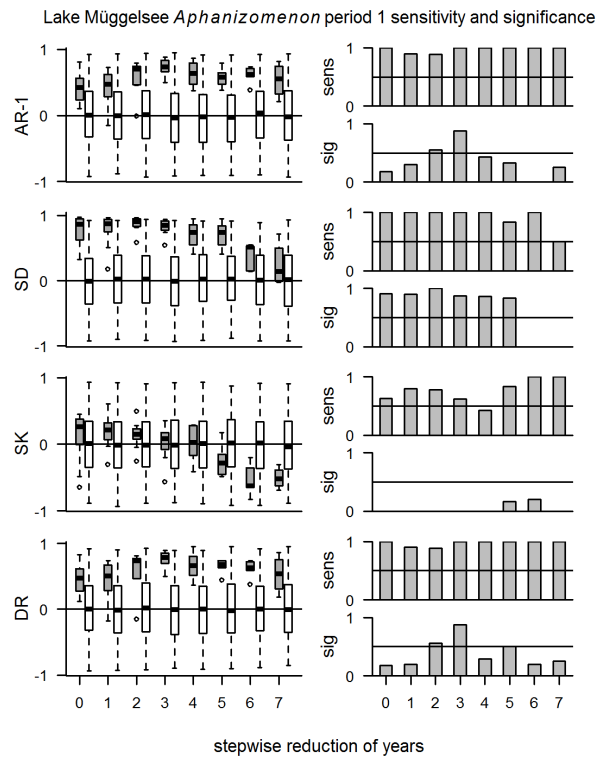
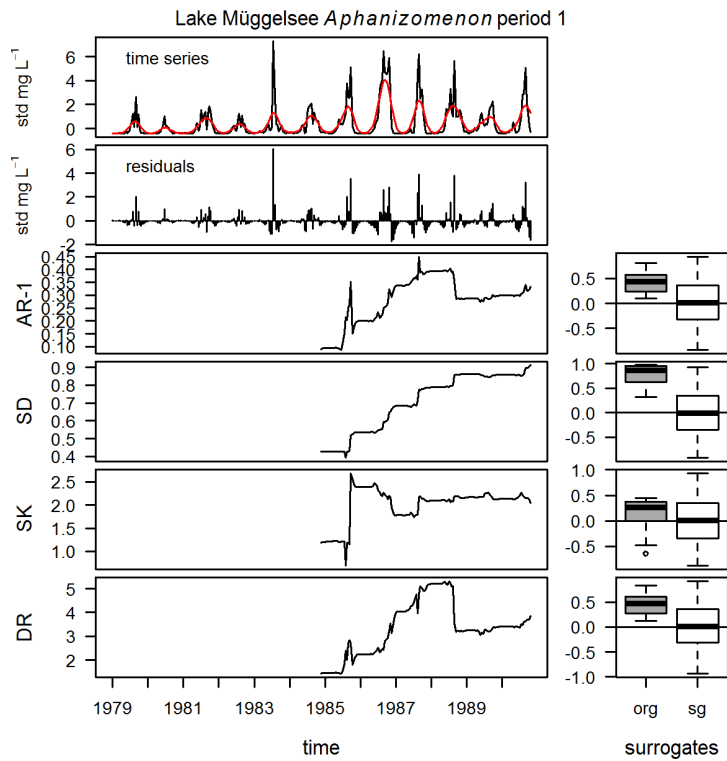
Right panels from top to bottom: ACF (upper panel), PACF (middle panel) and the periodogram (lower panel) of the seasonally adjusted time series up to the transition. Seasonal adjustment was performed by Gaussian filtering with a bandwidth fixed to the equivalent of one year worth of data (see also S6 for comparison of seasonal adjustment methods). Blue lines are 95% confidence intervals for a white noise process.

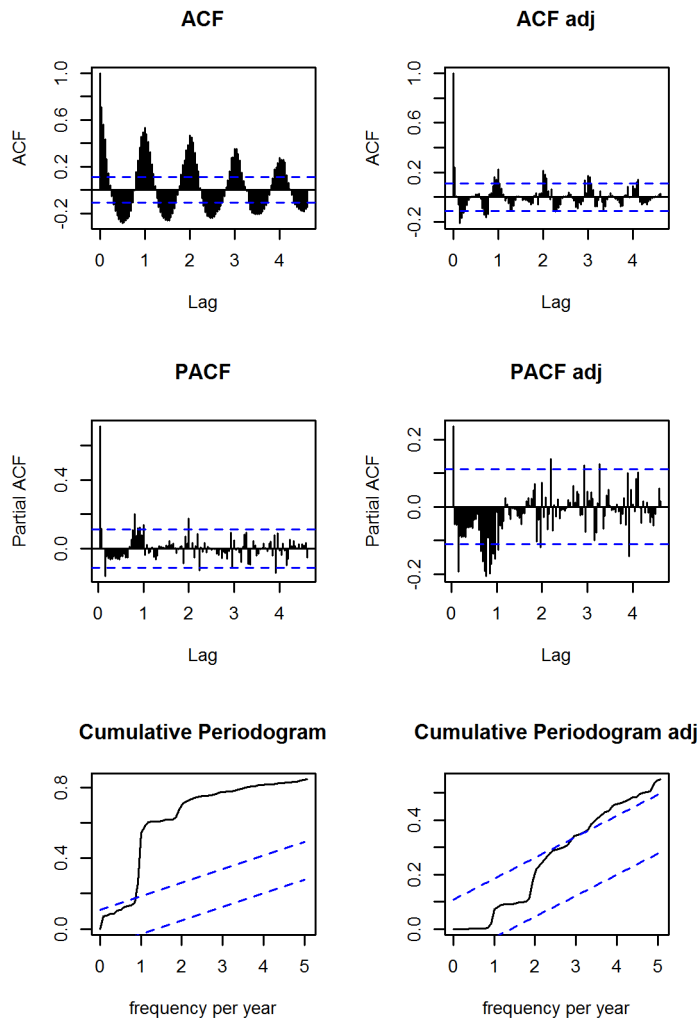
Lake Müggelsee: Phytoplankton biomass



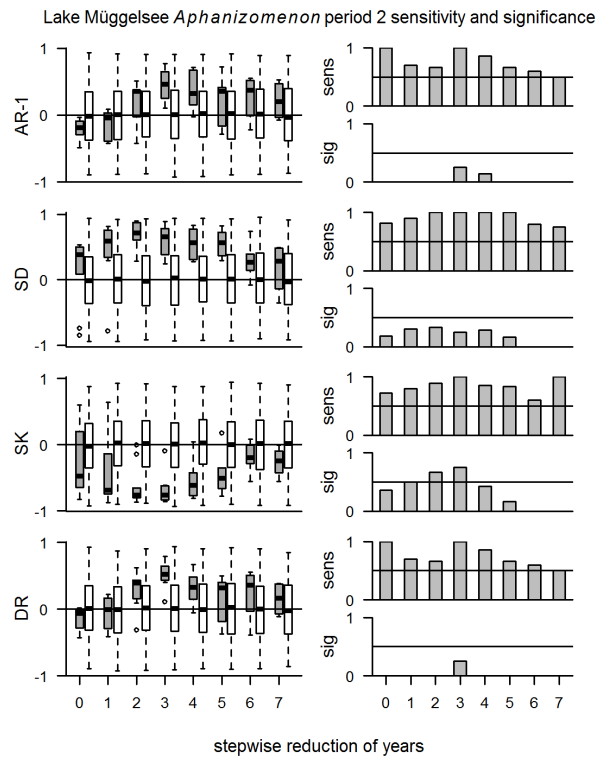
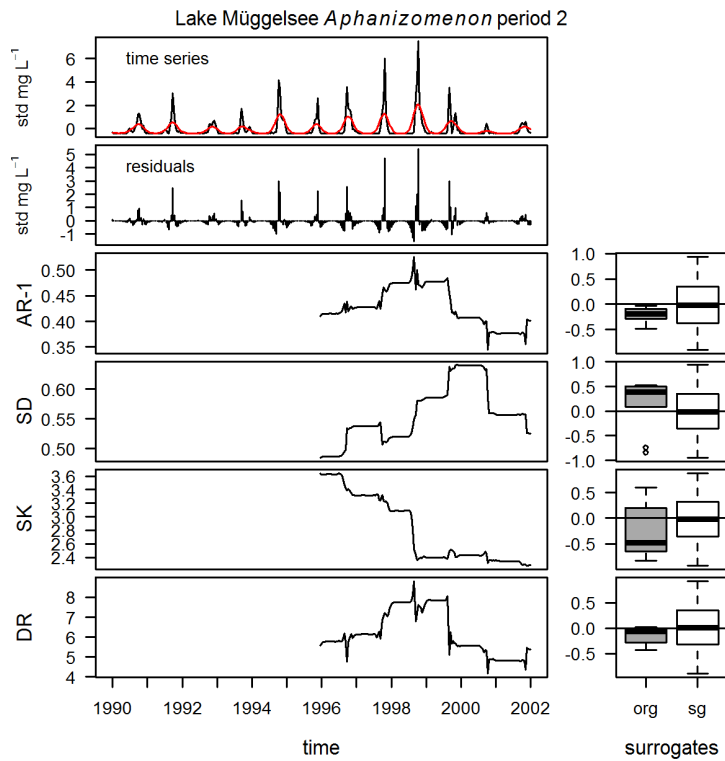


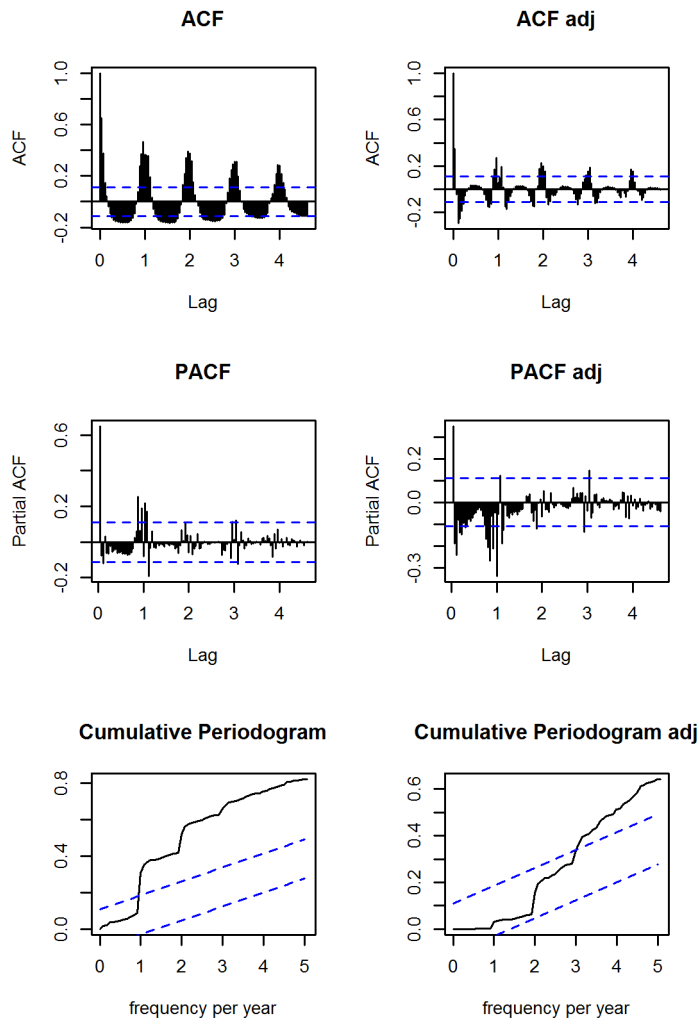
Lake Müggelsee: *Aphanizomenon* biomass period 1



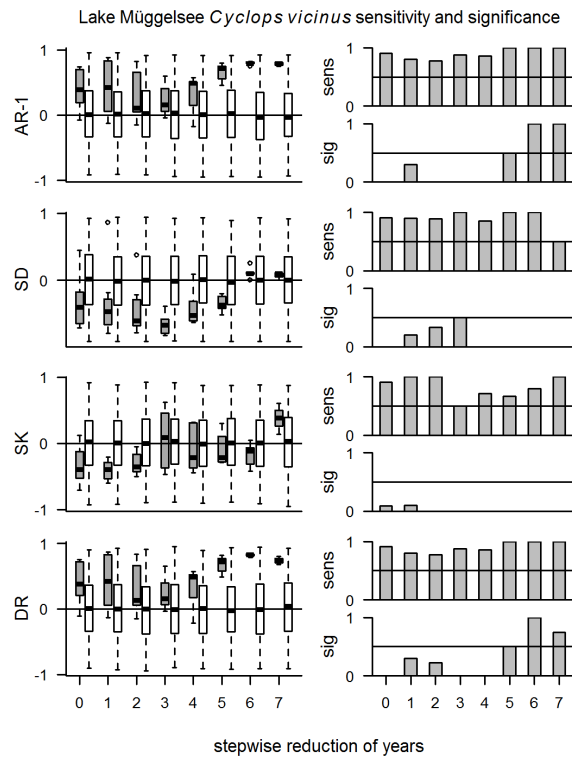
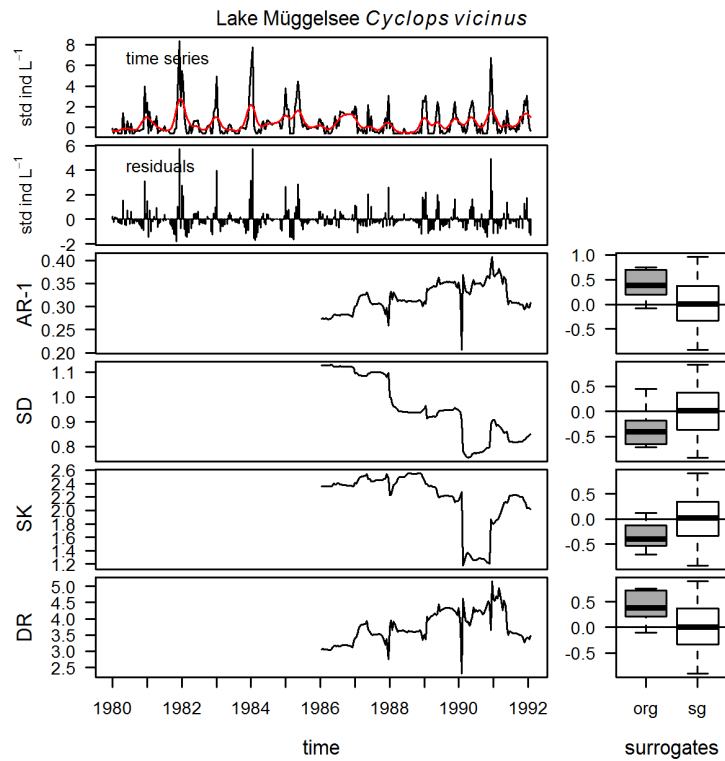


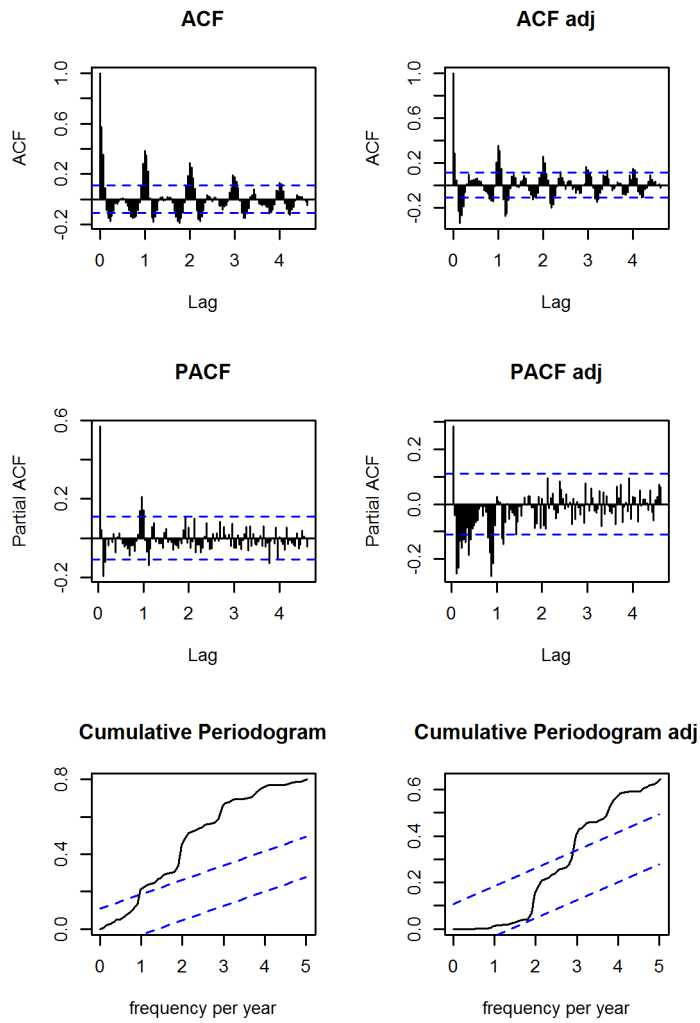
Lake Müggelsee: *Aphanizomenon* biomass period 2



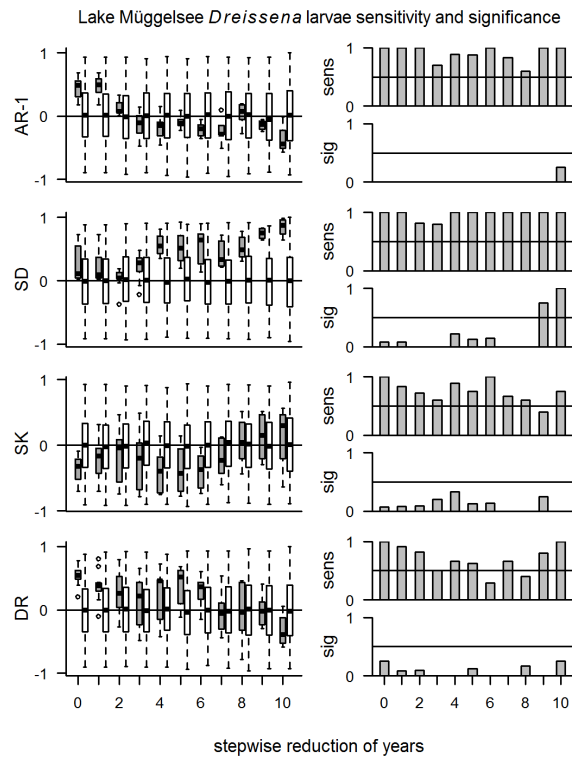
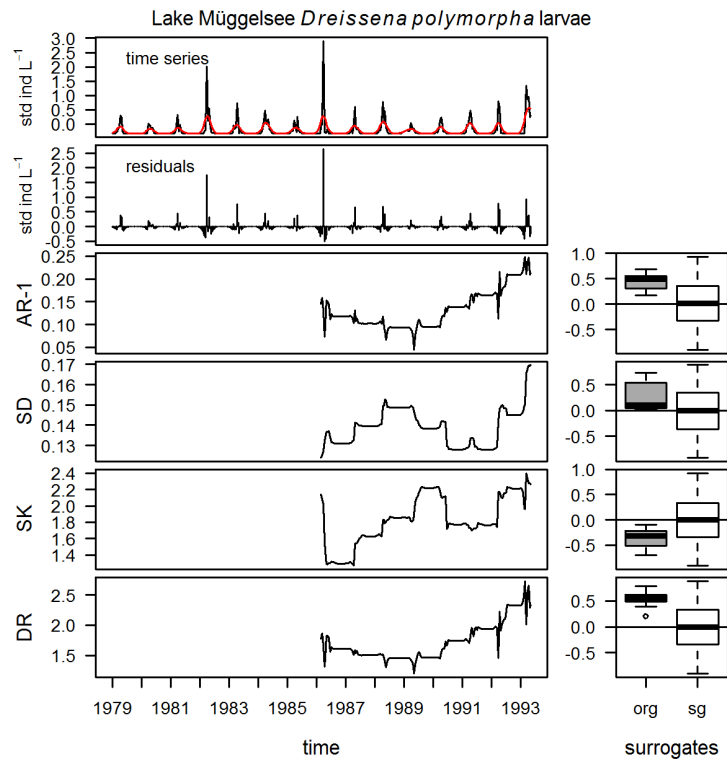


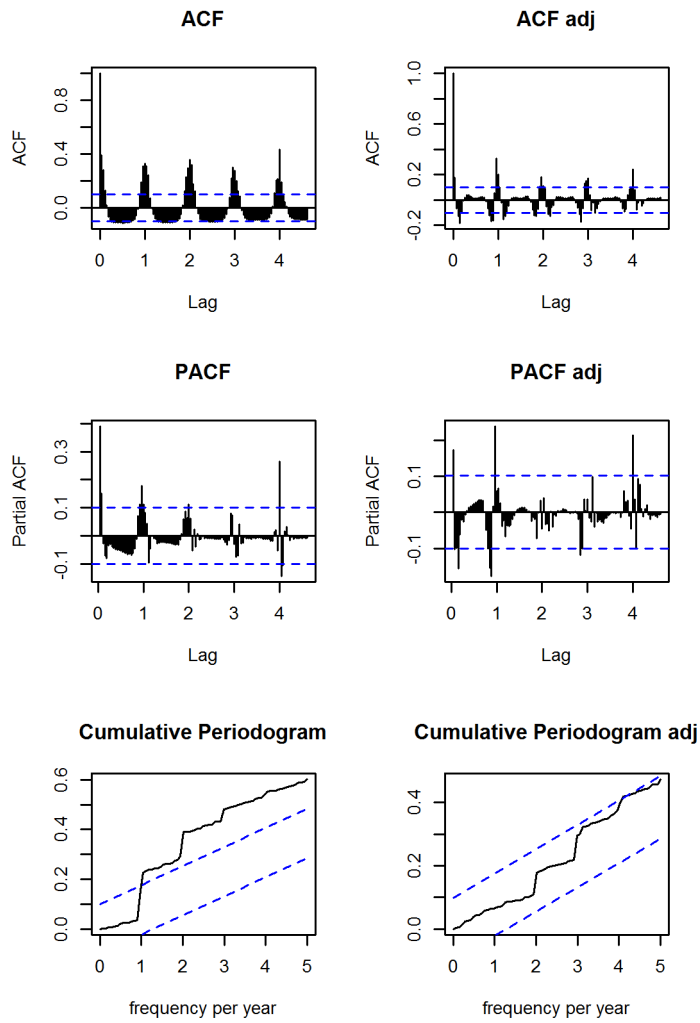
Lake Müggelsee: *Cyclops vicinus* abundance



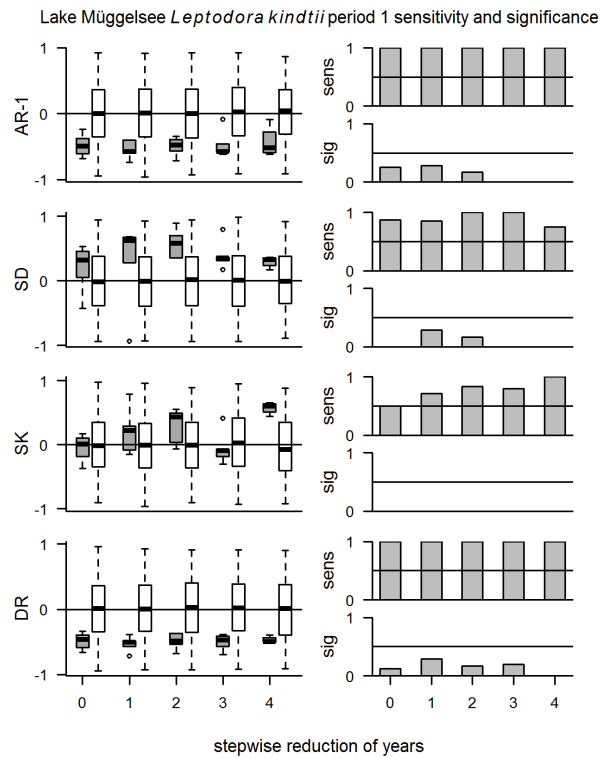
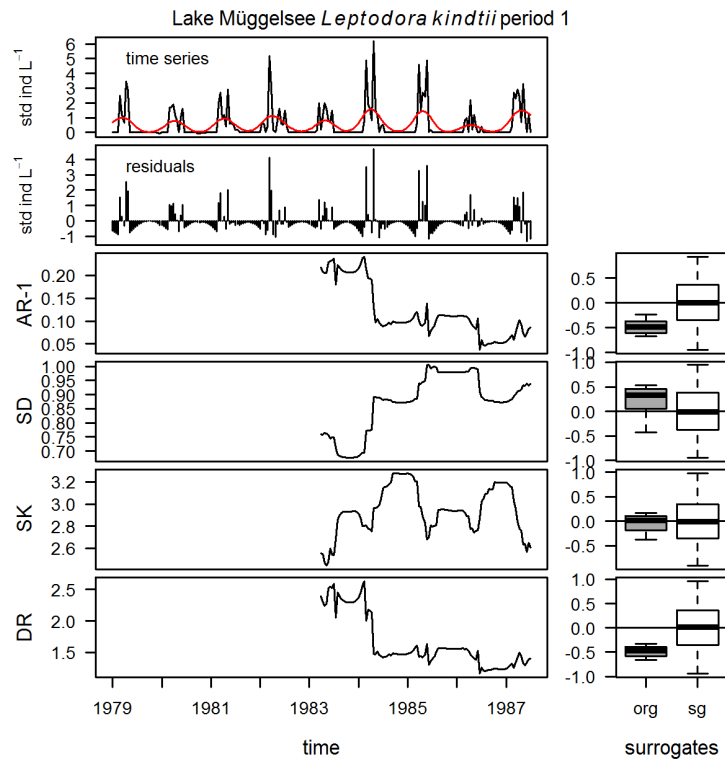


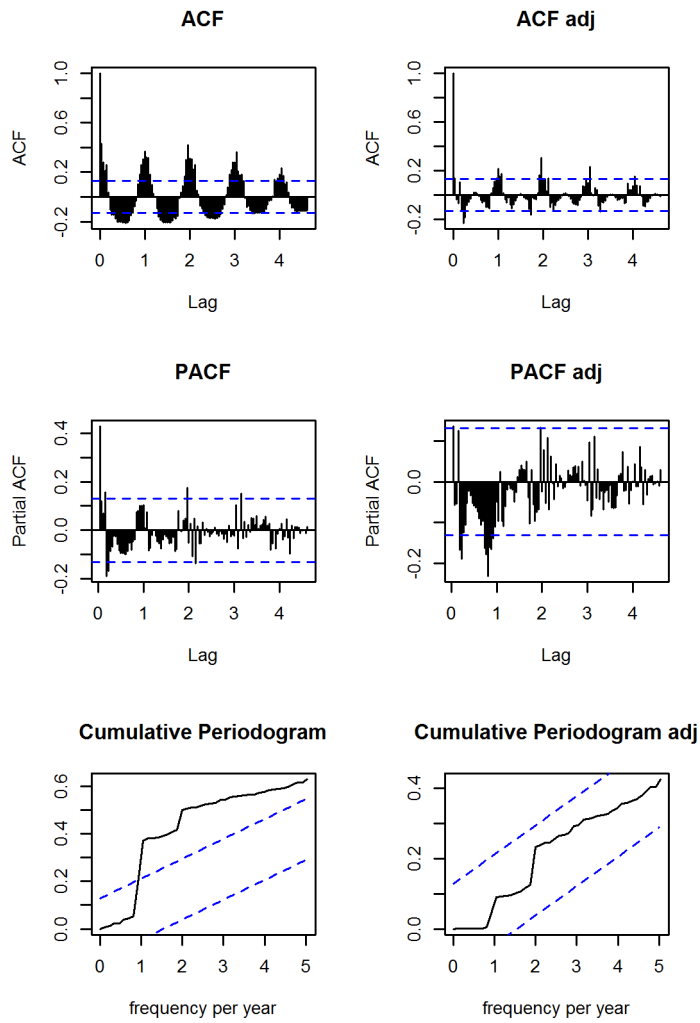
Lake Müggelsee: *Dreissena polymorpha* larvae abundance



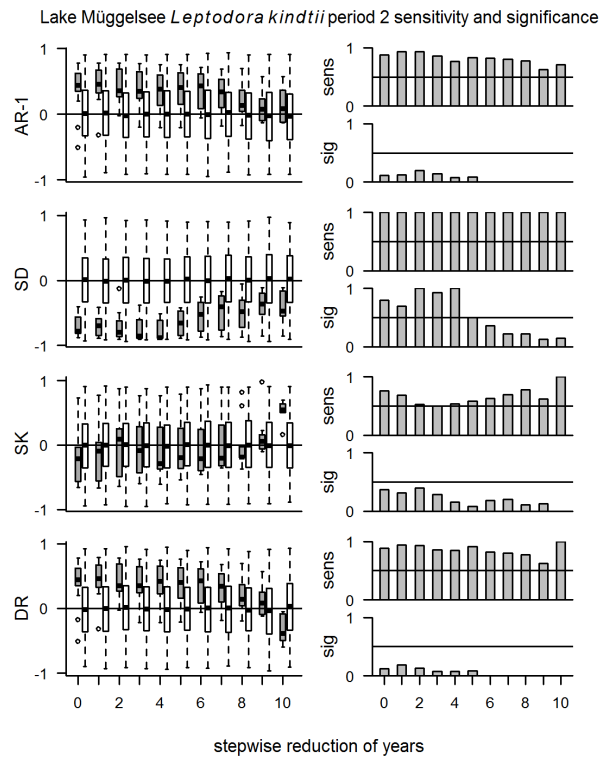
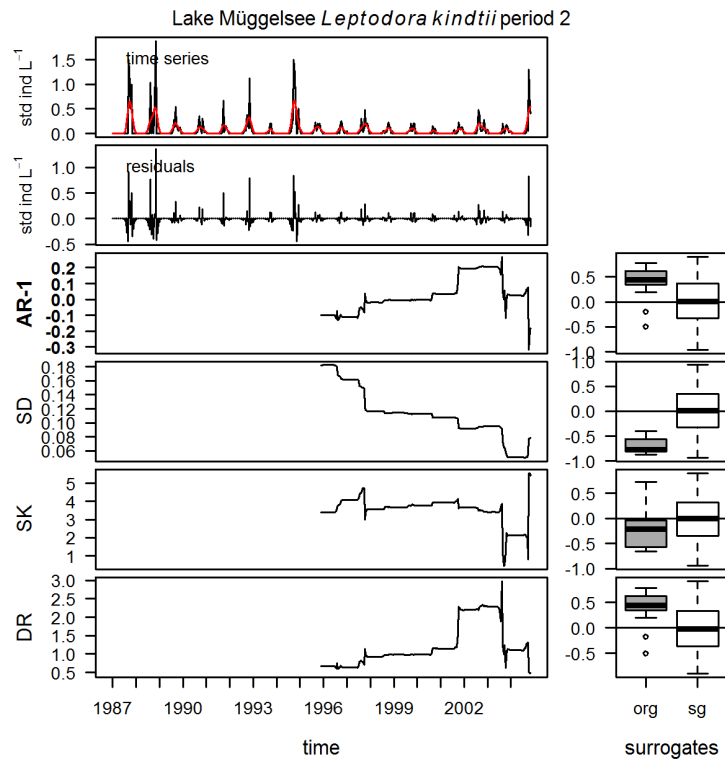


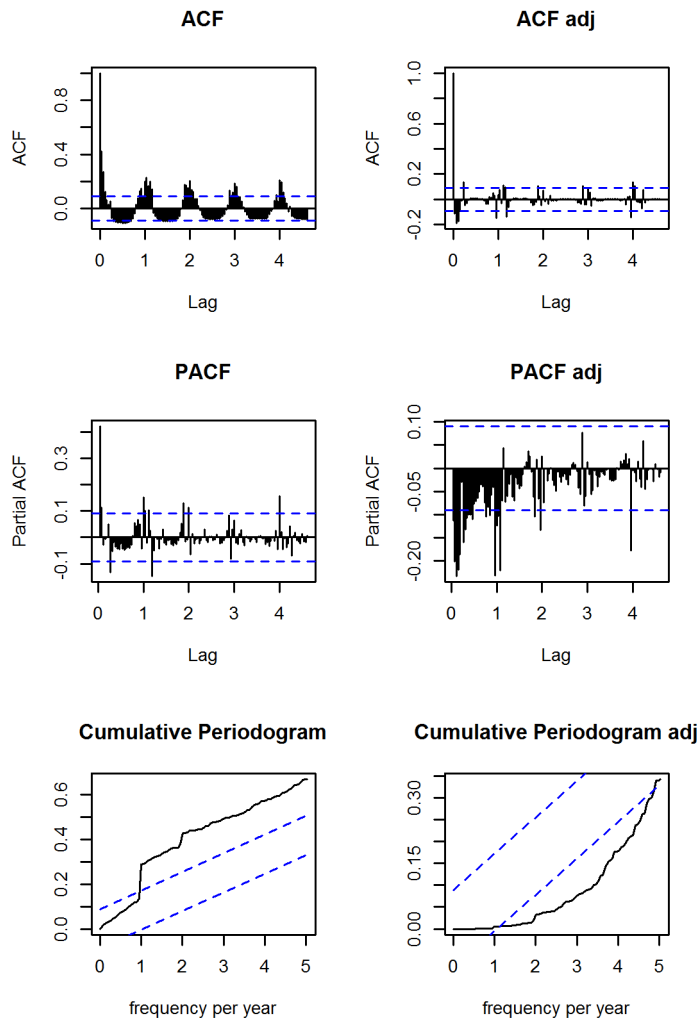
Lake Müggelsee: *Leptodora kindtii* abundance period 1



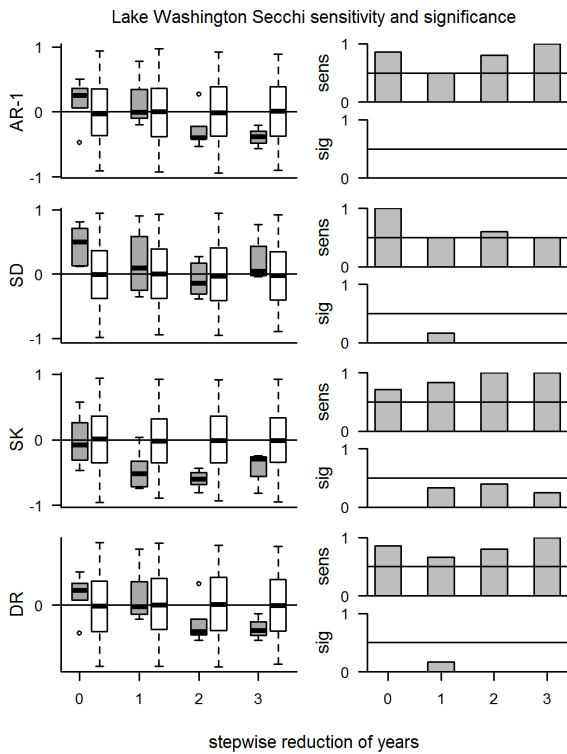
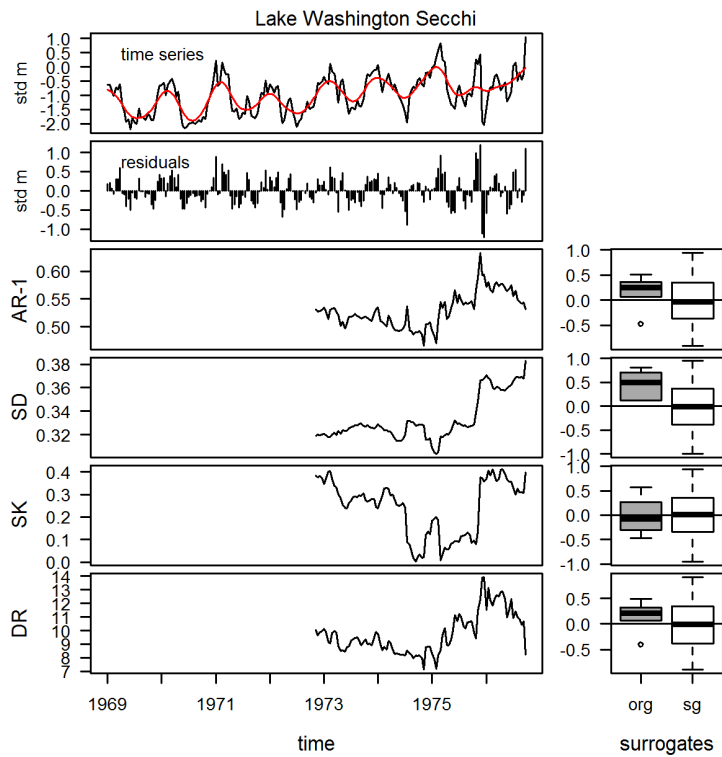


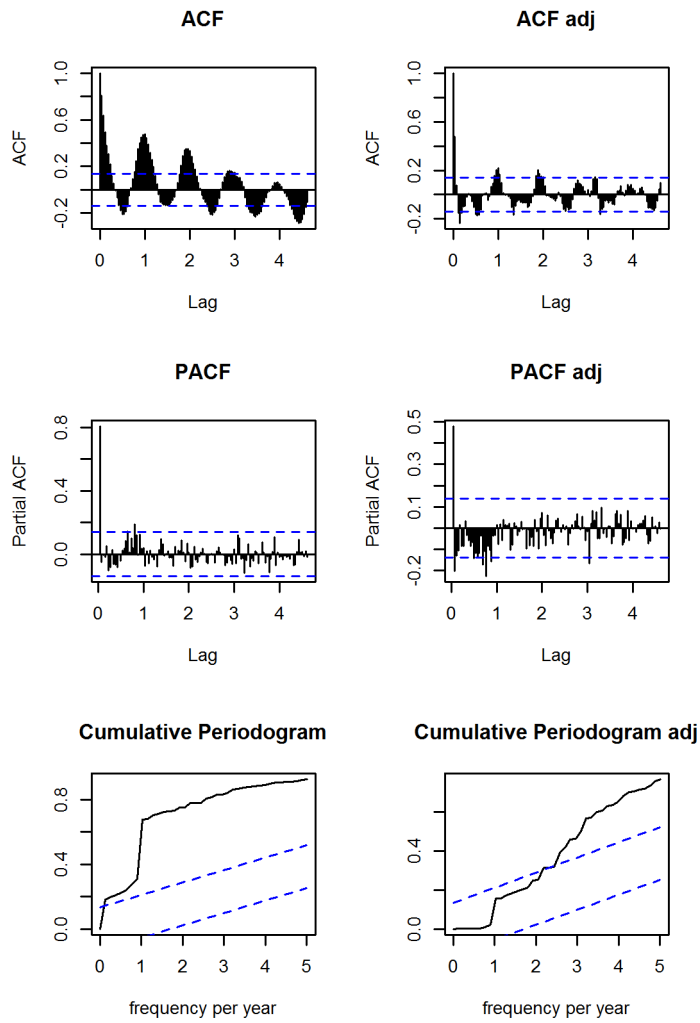
Lake Müggelsee: *Leptodora kindtii* abundance period 2



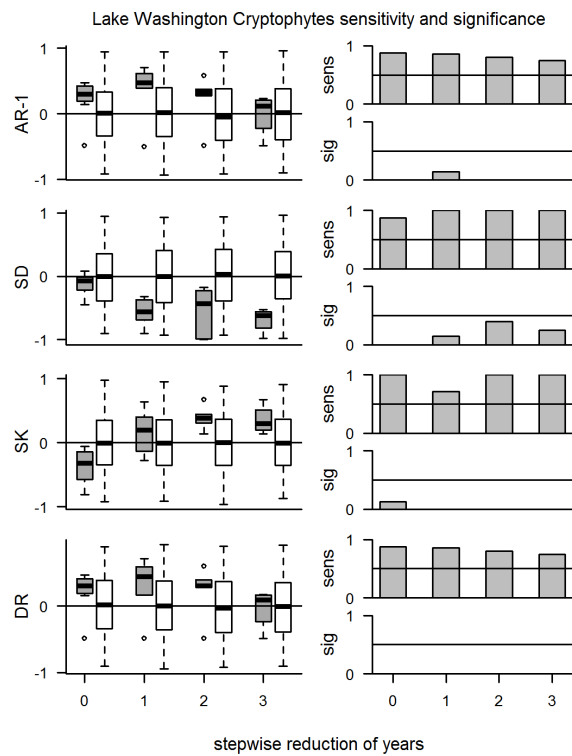
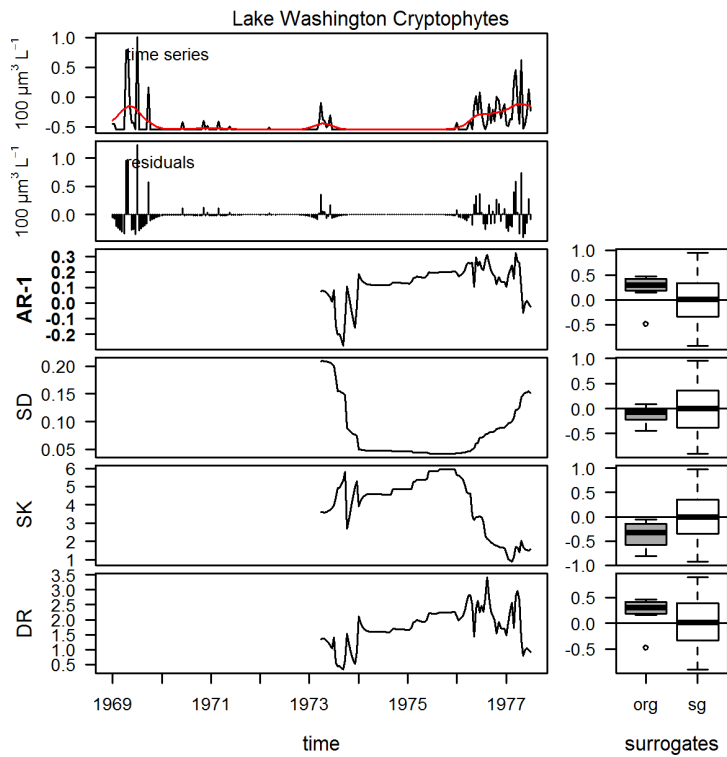


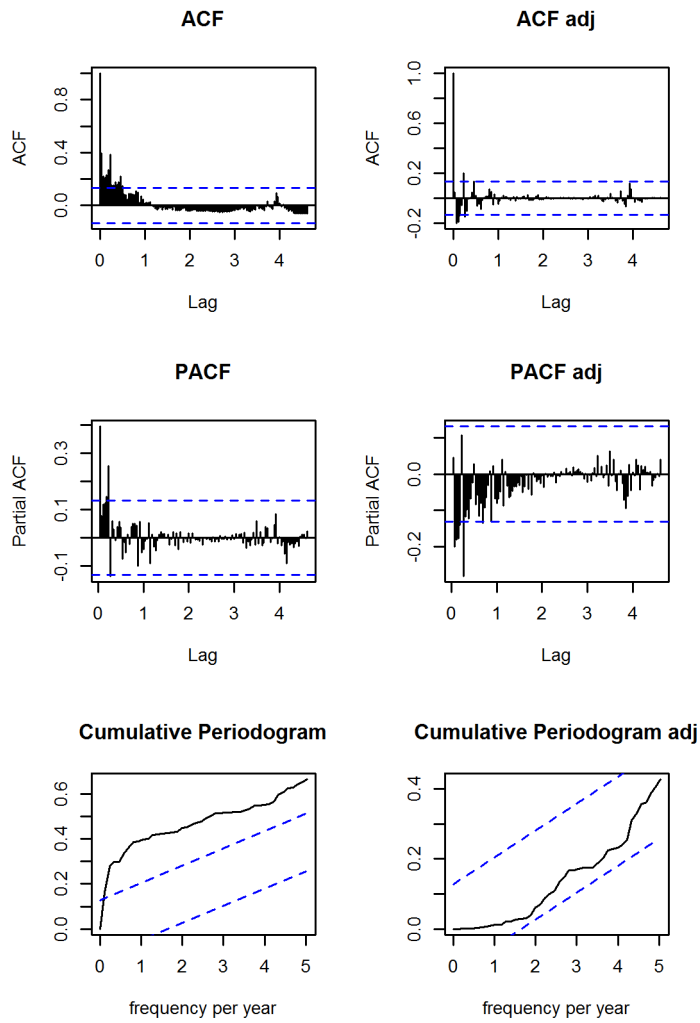
Lake Washington: Secchi depth



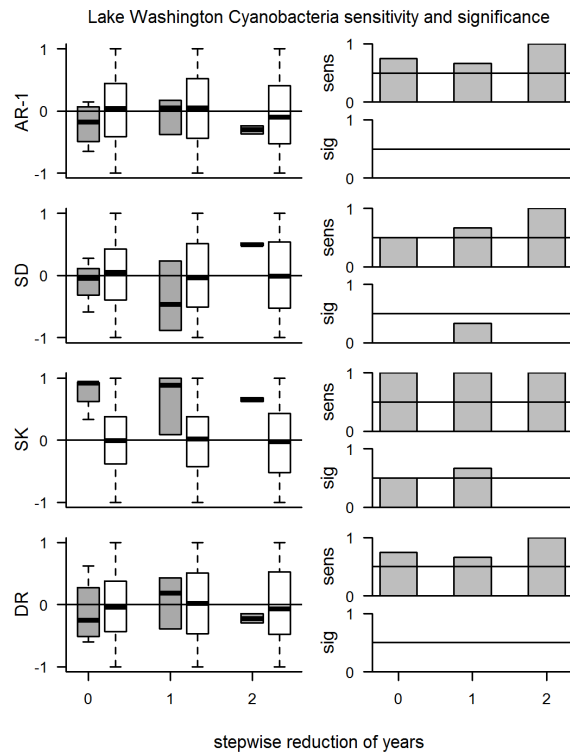
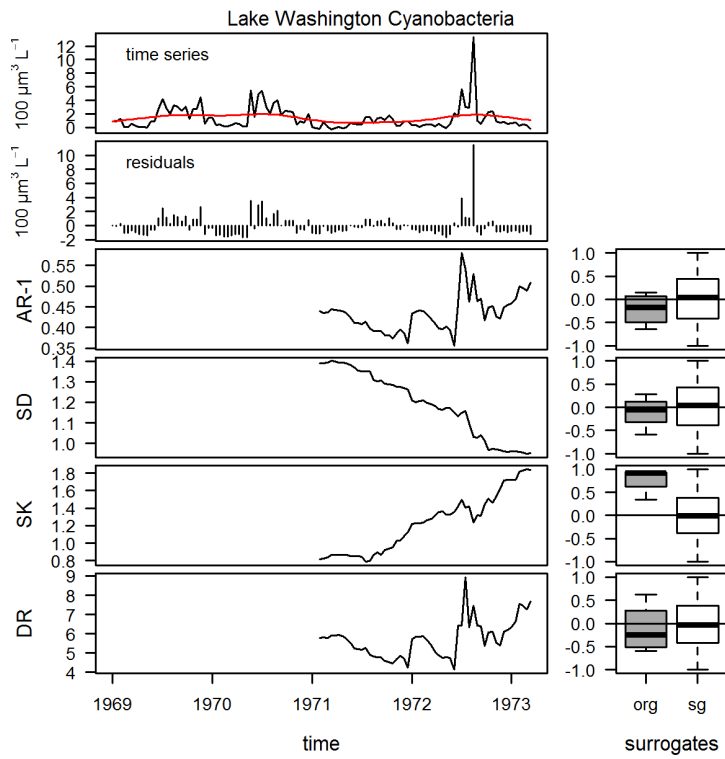


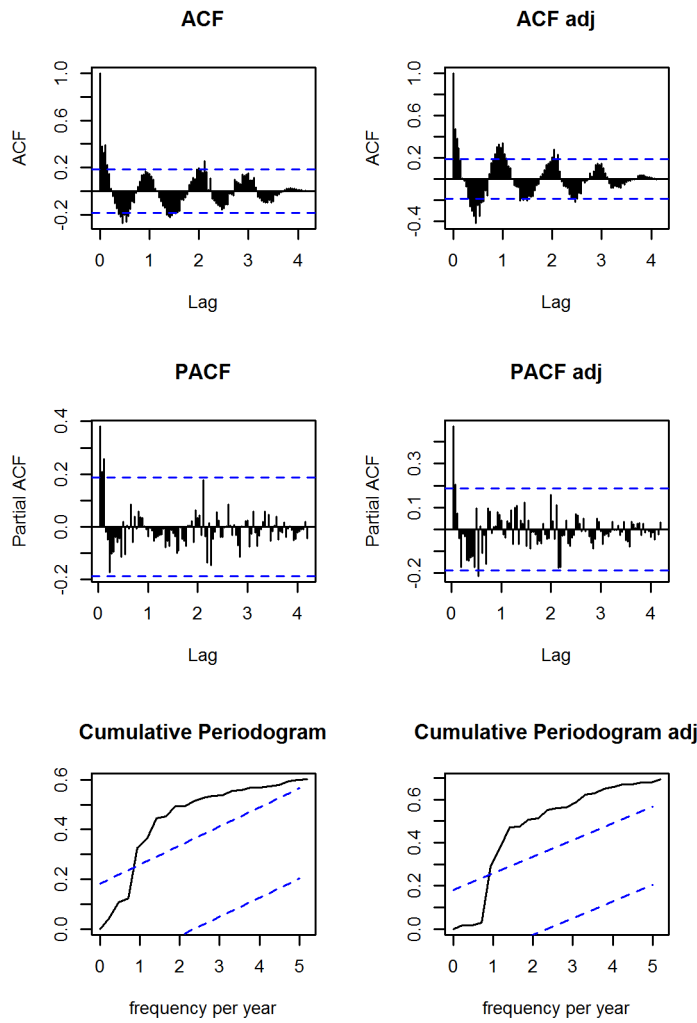
Lake Washington: Cryptophytes biomass



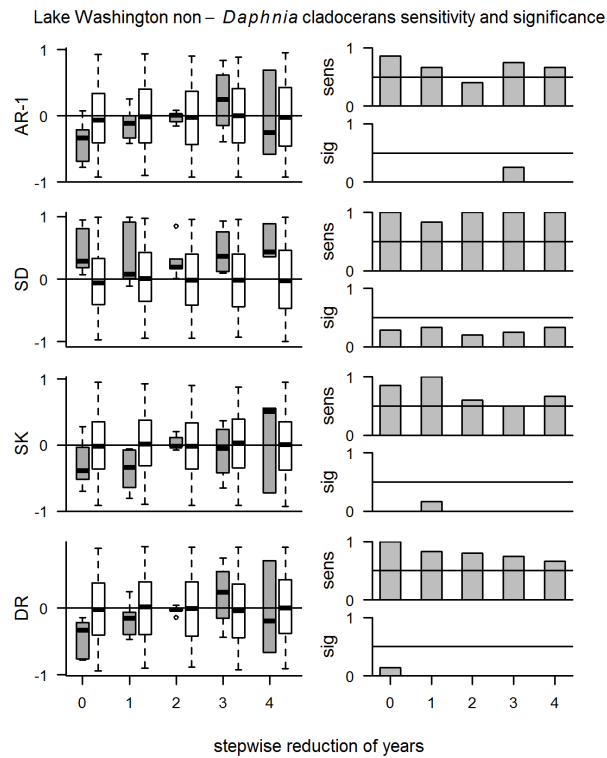
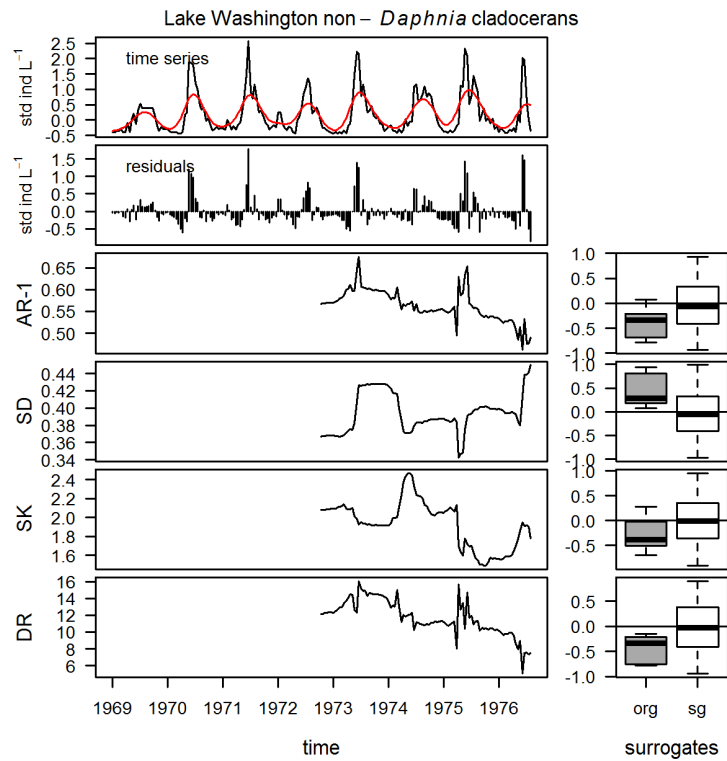


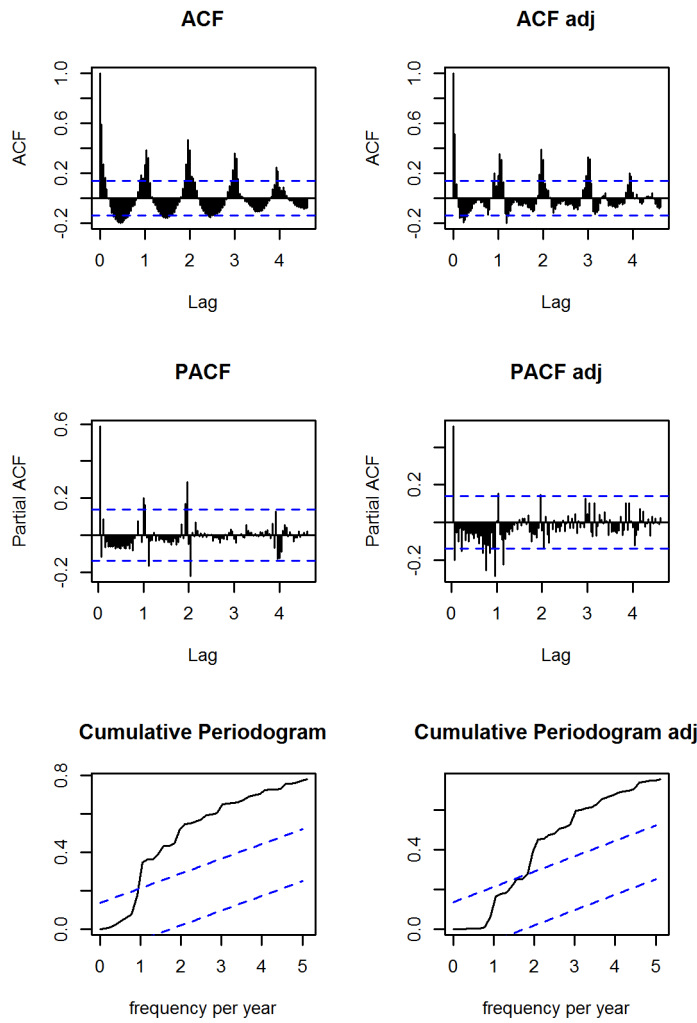
Lake Washington: Cyanobacteria biomass



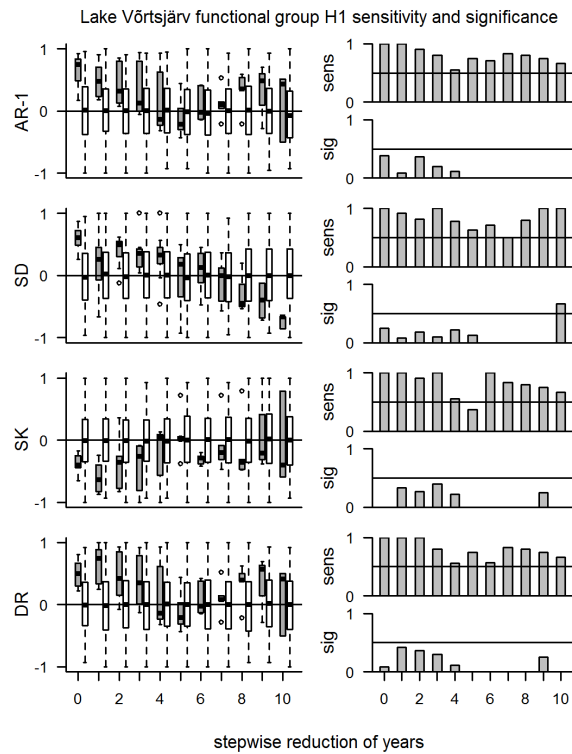
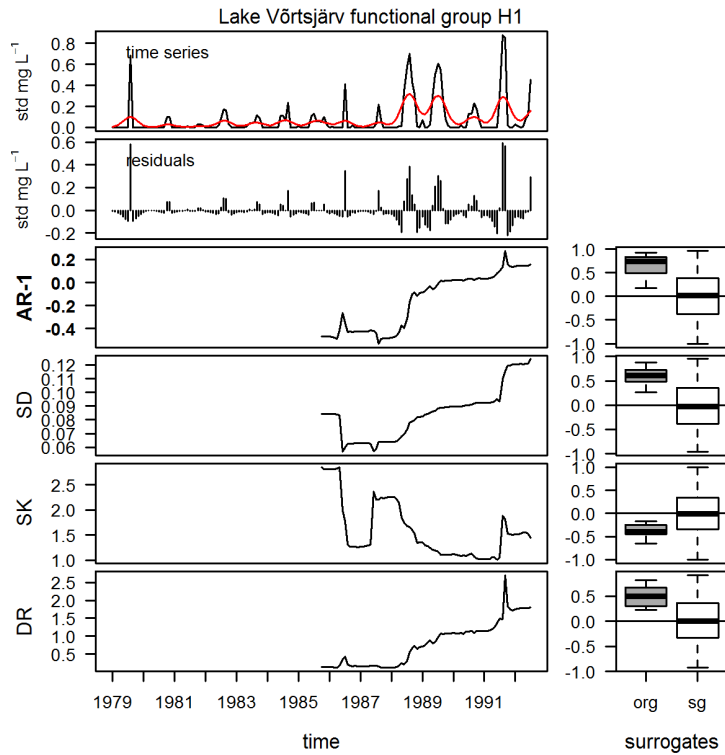


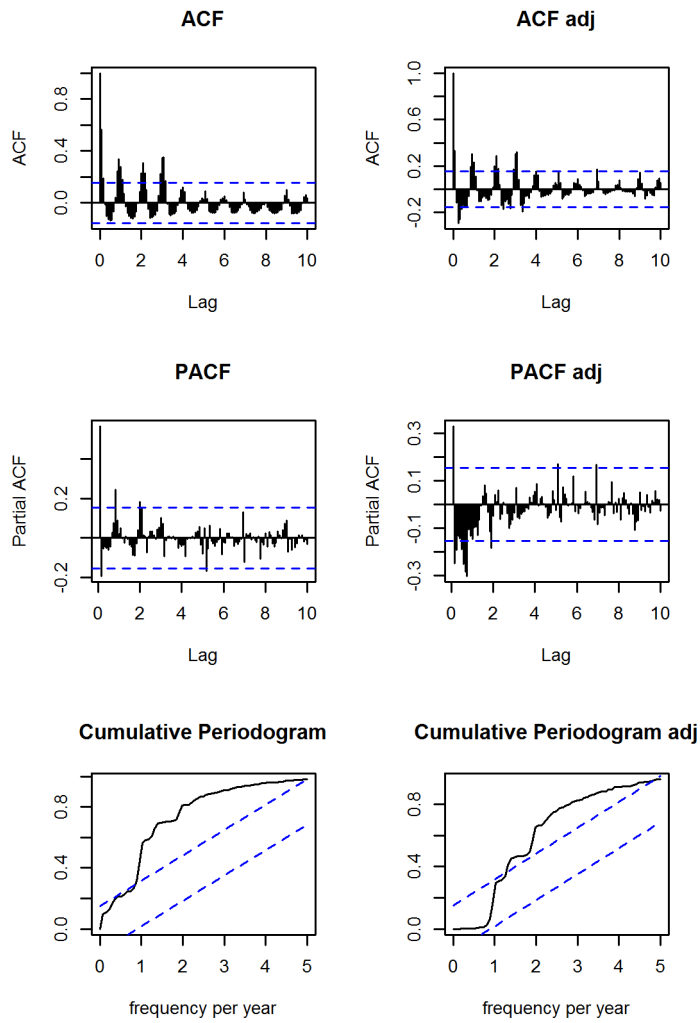
Lake Washington: Non-*Daphnia* Cladocera abundance



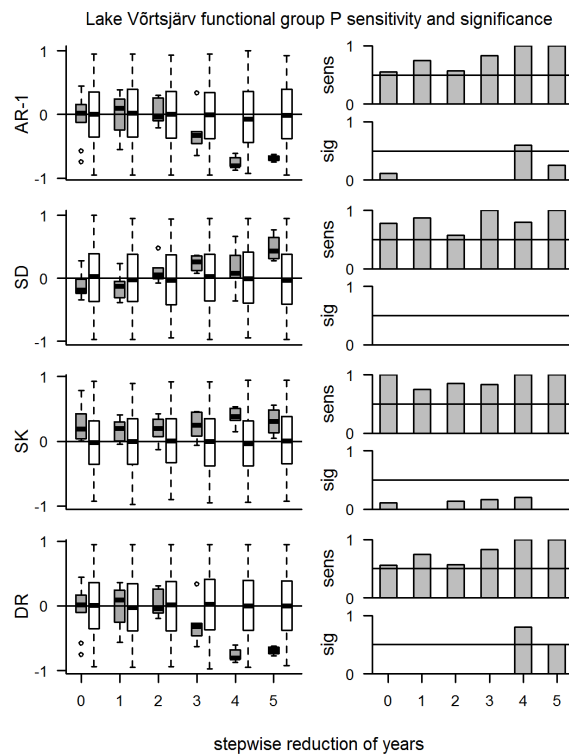
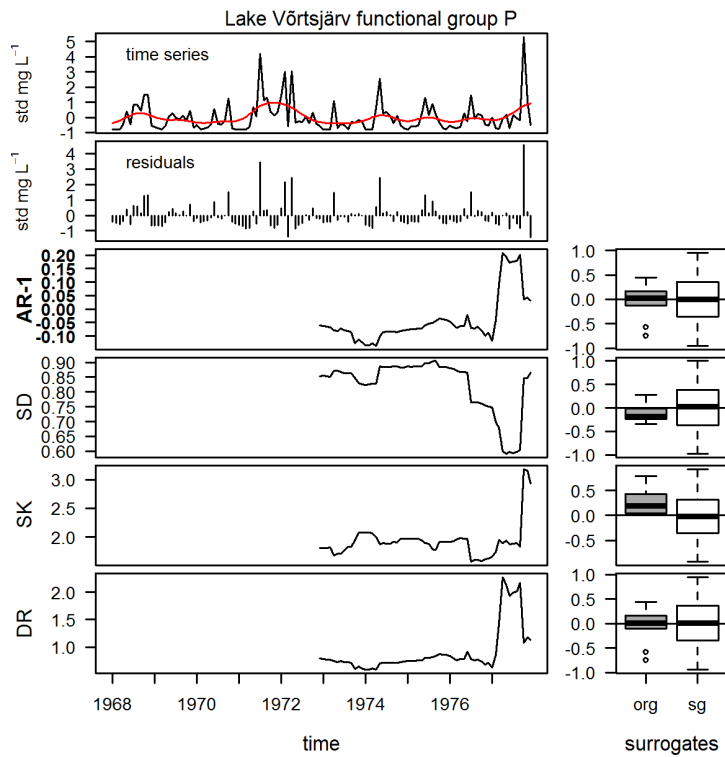


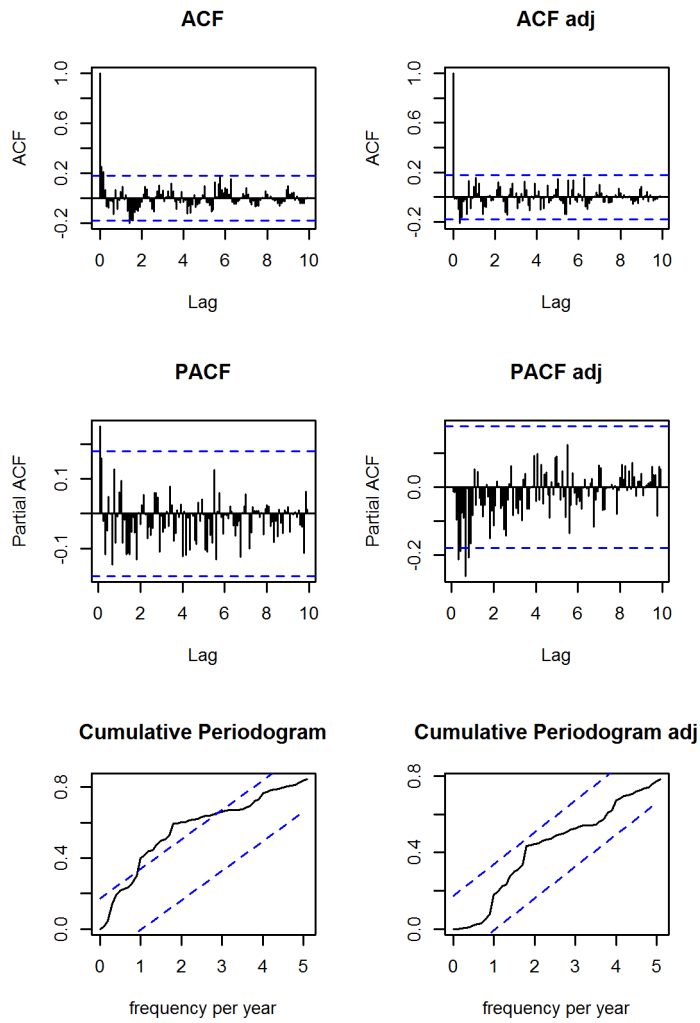
Lake Vörtsjärv: Functional group H1 biomass



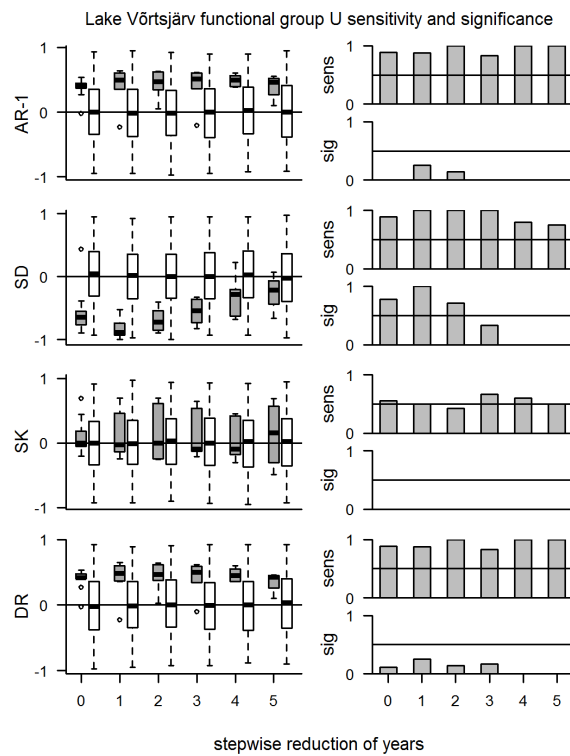
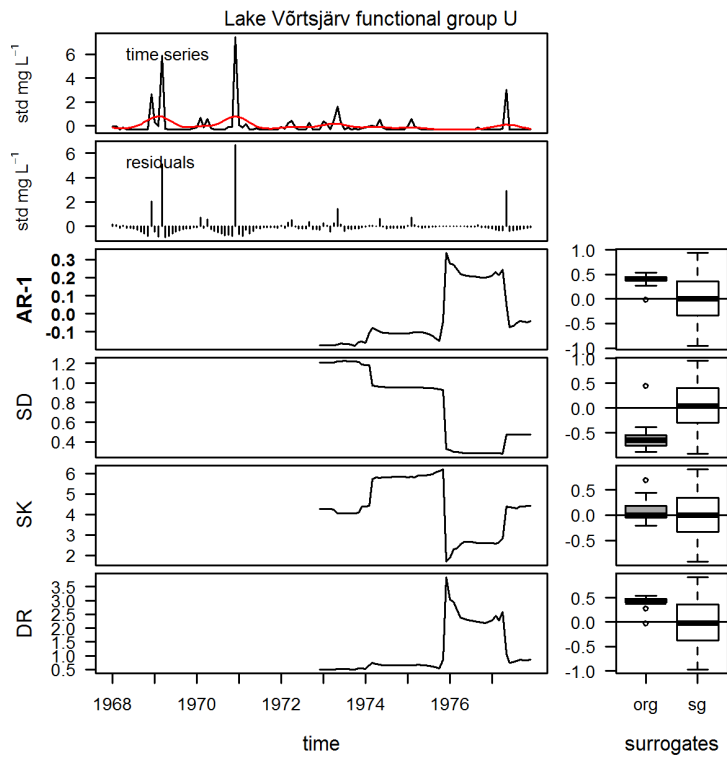


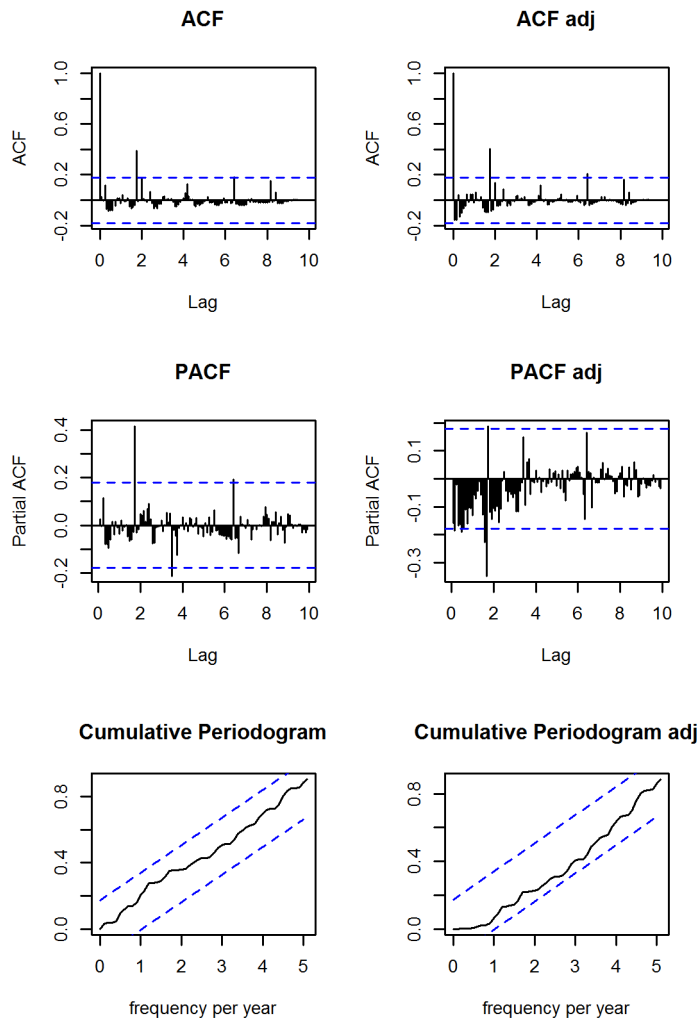
Lake Vörtsjärv: Functional group P biomass



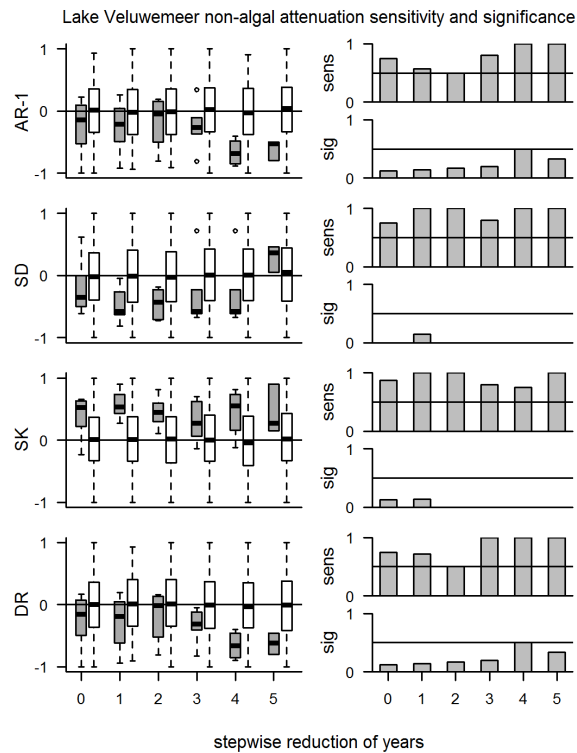
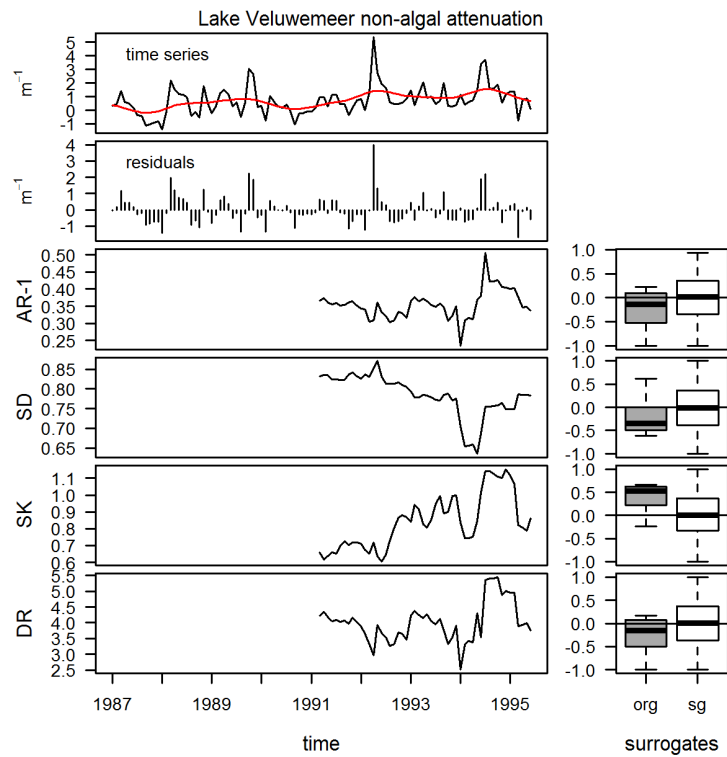


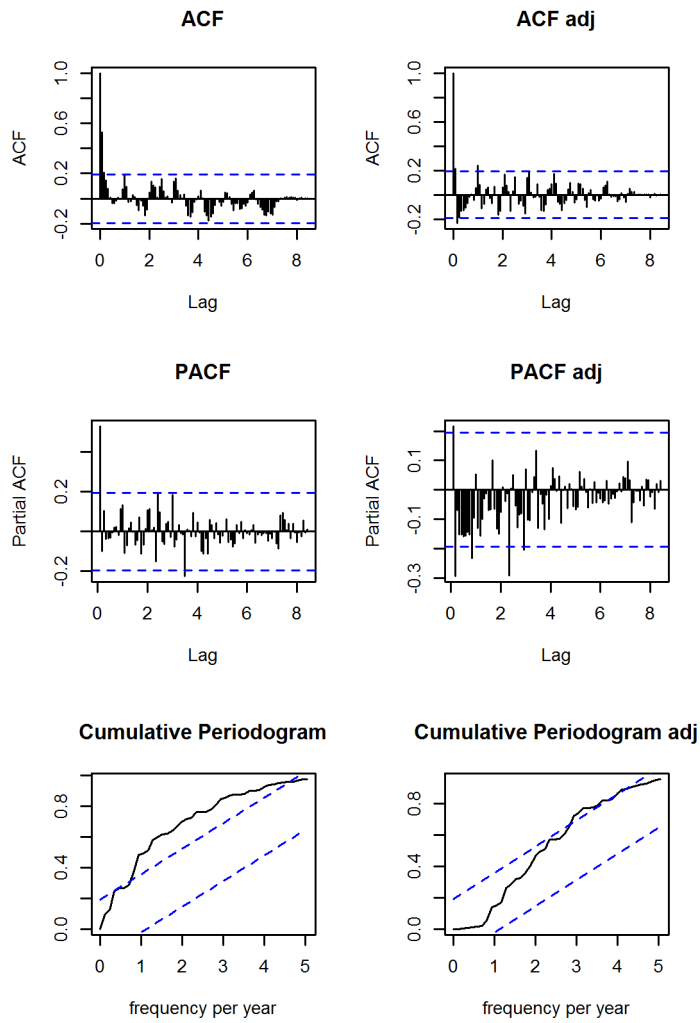
Lake Vörtsjärv: Functional group U biomass



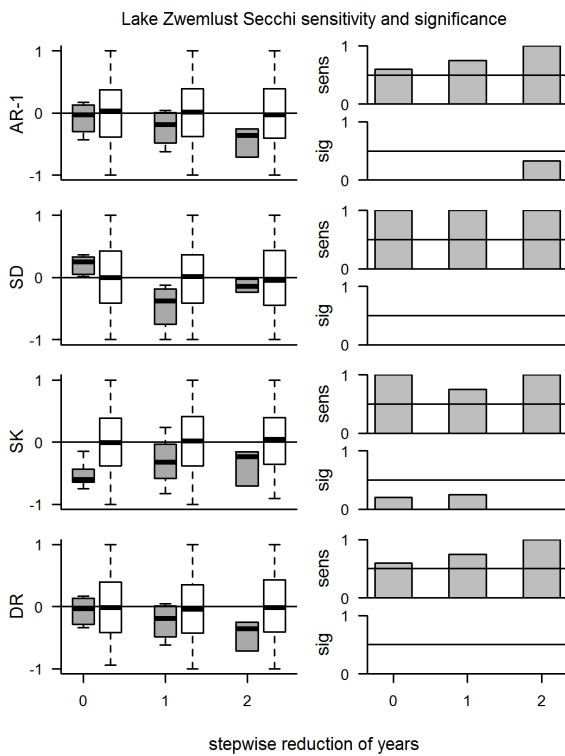
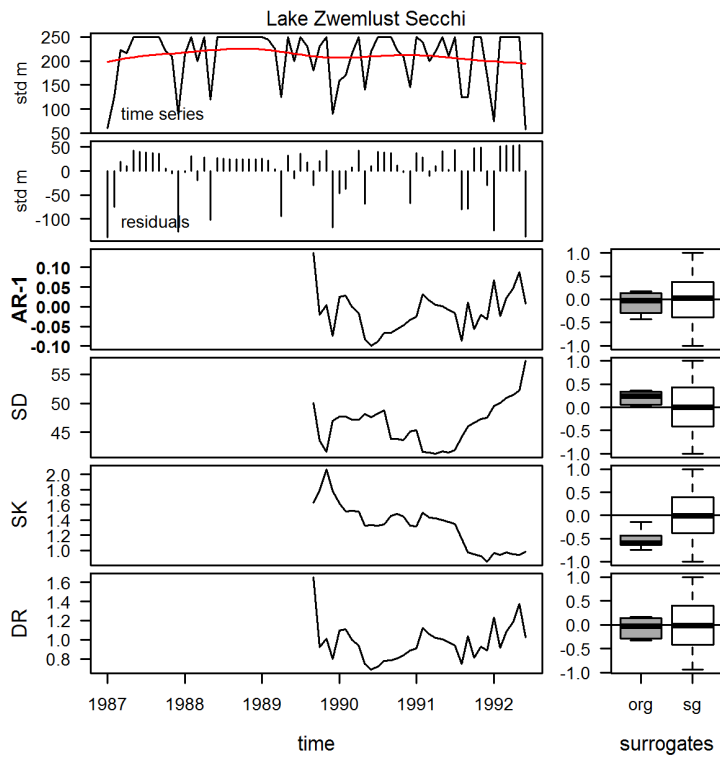


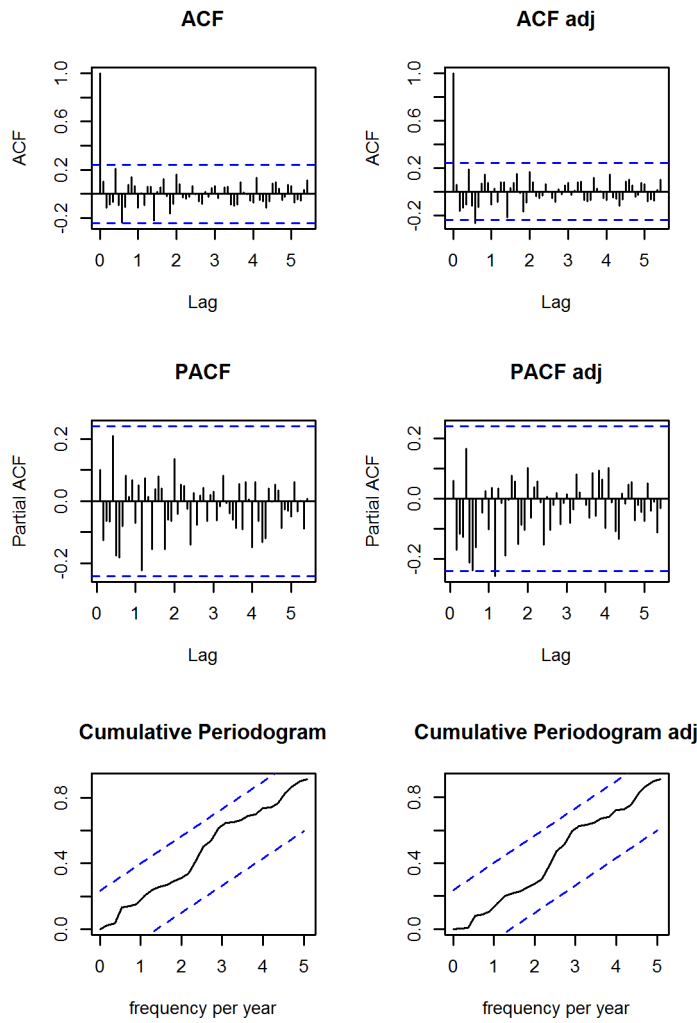
Lake Veluwemeer: Non-algal attenuation





Lake Zwemlust: Secchi depth





S4: Figure Robustness and significance of EWIs skewness and density ratio

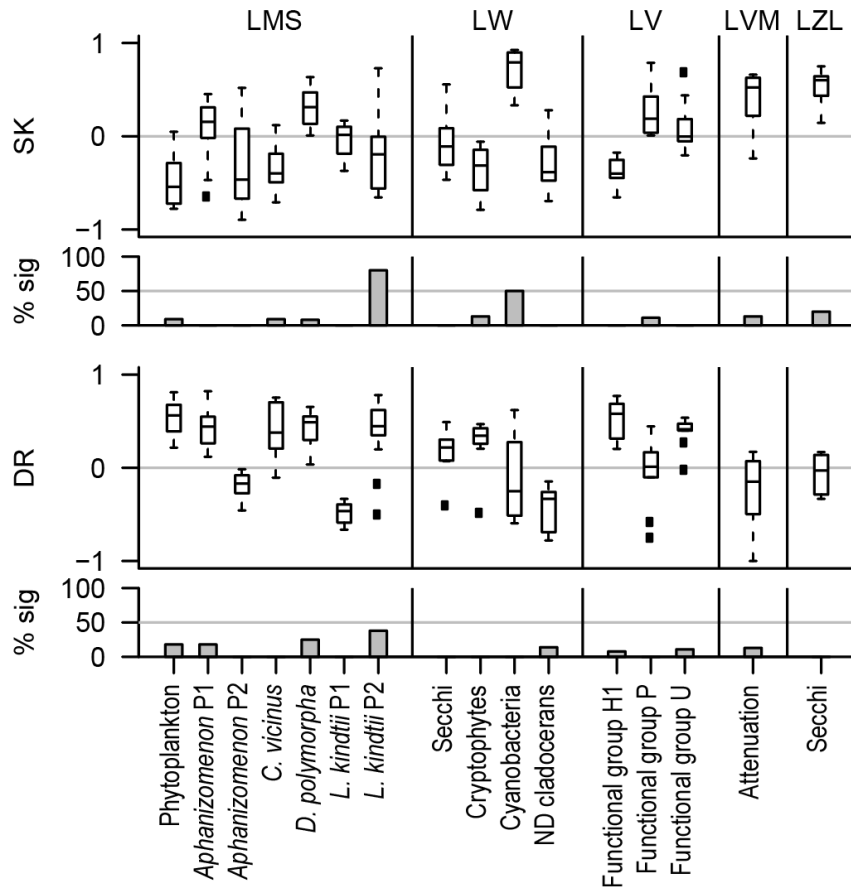


Figure S4: Mann-Kendall trend distributions for skewness and density ratio (panels) and per analysed state variable across all window sizes (boxplots) with the associated proportion of significant differences between trends based on data and surrogate time series (grey bar plot).

S5: Table S5: Early detection of early-warning indicators

Table S5: Early detection of EWIs expressed in number of years before the actual shift that each EWI would already have been detectable. Numbers indicate the maximum number of years deducted from the full pre-breakpoint time series before the theoretically expected trend disappeared (i.e. the median of the trend distribution changed sign). EWI trends were coded with ‘NA’ if they showed the opposite of the theoretically expected trends based on the full pre-breakpoint time series. Brackets denote AR-1 trends crossing zero on the scale.

Case Study	State Variable	AR-1	SD	SK	DR
LMS	Phytoplankton	3	6	4	3
	<i>Aphanizomenon</i> period 1	>7	>7	NA	>7
	<i>Aphanizomenon</i> period 2	NA	>7	>7	NA
	<i>Cyclops vicinus</i>	>7	NA	3	>7
	<i>Dreissena polymorpha</i> larvae	2	>10	NA	6
	<i>Leptodora kindtii</i> period 1	NA	>4	NA	NA
	<i>Leptodora kindtii</i> period 2	(>10)	NA	NA	9
LW	Secchi depth	0	1	NA	0
	Cryptophyceae	(>3)	NA	NA	>3
	Cyanophyceae	NA	NA	NA	NA
	Non- <i>Daphnia</i> cladocerans	NA	>4	3	NA
LV	H1	(3)	6	NA	3
	P	(1)	NA	>5	1
	U	(>5)	NA	1	>5
LVM	Non algal light attenuation	NA	NA	NA	NA
LZL	Secchi depth	(NA)	0	>3	NA

S6: Comparison of seasonal adjustment methods using PCLake modelling data

The aim of this supplementary material section is to test and compare several commonly used methods for their ability to remove trend and seasonal signals from empirical long-term data, a critical preprocessing step for the assessment of early-warning indicators. The following methods are tested: Piecewise linear regression, stable seasonal filter, Loess decomposition, moving averages, first differencing, and Gaussian filtering. For higher control over seasonal component and the regime-shift dynamics a modeled chlorophyll-a time series generated with PCLake is used.

PCLake is a well-tested and calibrated dynamic, mathematical model of eutrophication effects in shallow, non-stratifying lakes (23). PCLake is used to simulate and understand alternative stable states and hysteresis (24, 25). The model explicitly includes four trophic levels with several functional groups each. Moreover, the model is based on key ecological concepts including closed cycles of nutrients and matter, benthic-pelagic coupling, stoichiometry, food-web dynamics and trophic cascade. Due to its large number of state variables (>100) and parameters (>300), the model is relatively complex and hence provides an intermediary between output of minimal models which may underestimate complexity and observations in natural systems in which complexity and measurement errors are difficult to disentangle.

We run the model for a eutrophication scenario to simulate the typical loading history of many shallow lakes in the temperate zone during the second half of the twentieth century. Therefore we used the default parameter settings of an average temperate lake as calibrated by Janse *et al.* (24). Additionally, we added stochasticity to all differential equations of PCLake to mimic the natural variability found in lakes. Therefore we used the ‘dwienner’ function in MATLAB which simulates white noise following the Wiener process (26). Accordingly, the change in a state variable S over time depends on the change due source and sink terms such as growth and mortality ($f(t)dt$) and on natural stochasticity calculated with a normal distribution ($g(t) \cdot DW$):

$$\frac{dS}{dt} = f(t)dt + g(t)DW$$

In our run, the noise is calculated with a standard deviation equal to 1% of the current state variable S . From this run we obtained the seasonally forced chlorophyll-a concentration with natural stochasticity to characterize the state of the lake ecosystem; chlorophyll-a is one of the

most common proxies for water quality used by ecosystem managers. This time series is then used to assess the timing of the breakpoint (when does the regime shift take place) and the pre-breakpoint time series is then used to test 6 seasonal adjustment methods and their effect on the resulting early warning indicator (EWI) trends. Testing the seasonal adjustment methods on PCLake model data rather than one of the real world time series presented in the main body of this manuscript is advantageous as the model data contain a true critical transition and therefore allow an assessment of a) how well different methods remove seasonal signals, and b) how different methods affect the EWI trend evolutions. In the following we present the results for the 6 different methods. The method performance is compared with the help of estimated ARIMA parameter for autoregression, moving average and season using the function `auto.arima` from the R-package `forecast` (27, 28), visual inspection of the autocorrelation function (ACF) and partial autocorrelation function (PACF), and most importantly the cumulative periodogram. In the cumulative periodogram, successful removal of the seasonal component from the signal should lead to random energy levels at the frequency content of one and two years per cycle.

S6.1 Original Chlorophyll-a Time Courses under Increasing Eutrophication

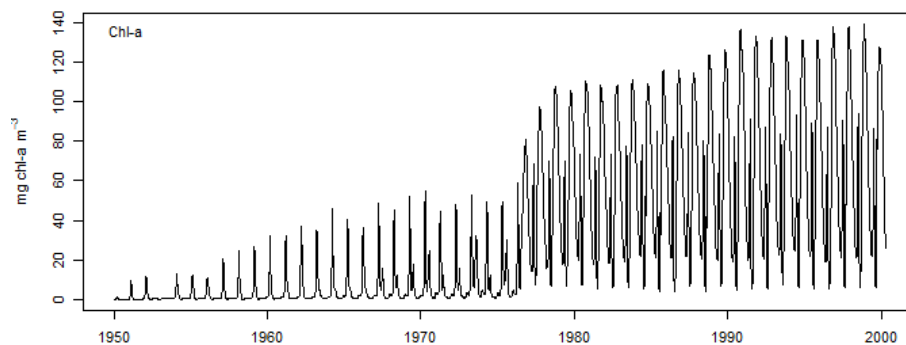


Figure S 6.1: Weekly chlorophyll-a data of 50 years model simulation under gradual eutrophication. The breakpoint from a clear to a turbid water system occurs in the year 1976.

S6.2 Original and Log-Transformed Chlorophyll-a Time Courses up to the 1976 Breakpoint

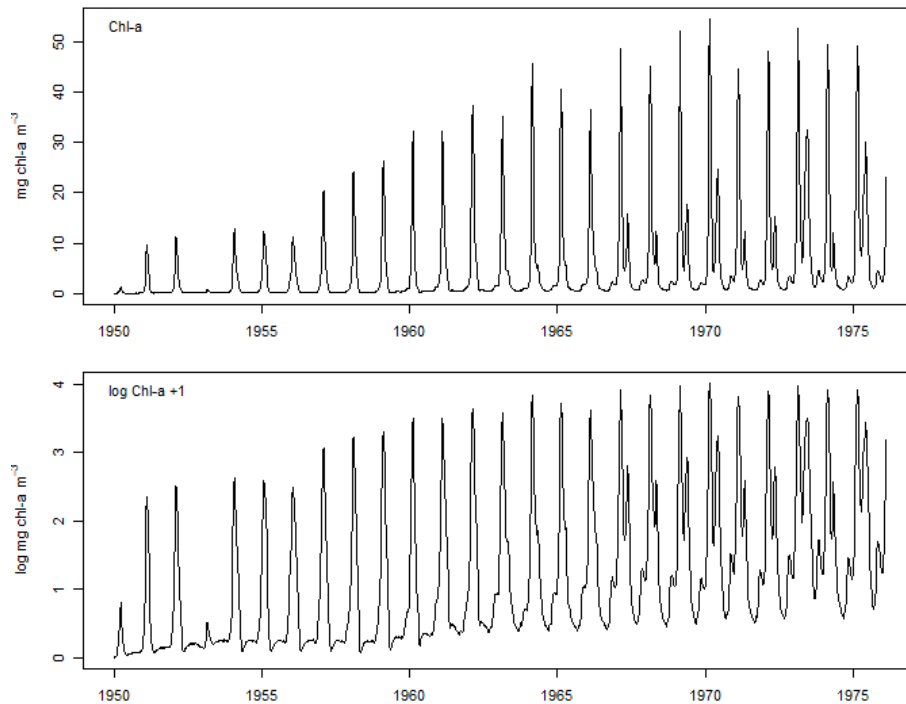


Figure S 6.1: Weekly original (upper panel) and log (chl-a+1) transformed (lower panel) time courses up to the breakpoint in 1976

S6.2.1 Seasonal Signal in the Original Time Series Assessed by Function `auto.arima` in R

ARIMA(2,1,2)(0,0,1)[52]

Coefficients:	ar1	ar2	ma1	ma2	sma1
	1.4542	-0.7148	-0.0088	-0.4954	0.3835
s.e.	0.0490	0.0237	0.0755	0.0692	0.0244

σ^2 estimated as 1.253: log likelihood=-2082.94

AIC=4177.87 AICc=4177.93 BIC=4209.15

S6.2.2 Assessing Seasonal Signals by Acf, Pacf and Periodogram of Original Time Series

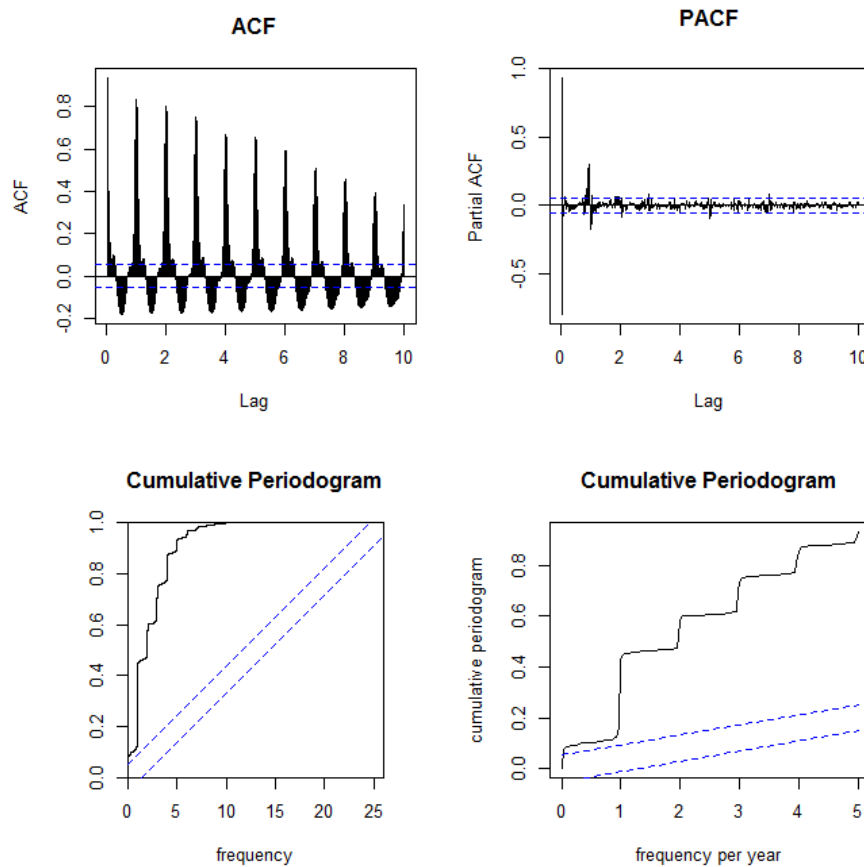


Figure S6.3: Assessment of the seasonal signal in the time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.2.3 Seasonal Signal in the Log Transformed Time Series Assessed by `auto.arima` in R

Log transformation of the original time series changes the amplitudes but not the periodicity or autocorrelation structure. When seasonal adjustment methods fail to take into account an interaction between time and amplitude of seasonal signal, it can be better to stabilize the seasonal amplitudes by log-transformation.

ARIMA(2,1,2)(0,0,1)[52]

Coefficients:	ar1	ar2	ma1	ma2	sma1
	1.5861	-0.7429	-0.1769	-0.2130	0.2469
s.e.	0.0353	0.0277	0.0458	0.0386	0.0245

σ^2 estimated as 0.004589: log likelihood=1728.55

AIC=-3445.1 AICc=-3445.04 BIC=-3413.82

S6.2.4 Assessing Seasonal Signals by Acf, Pacf and Periodogram of Log Transformed Time Series

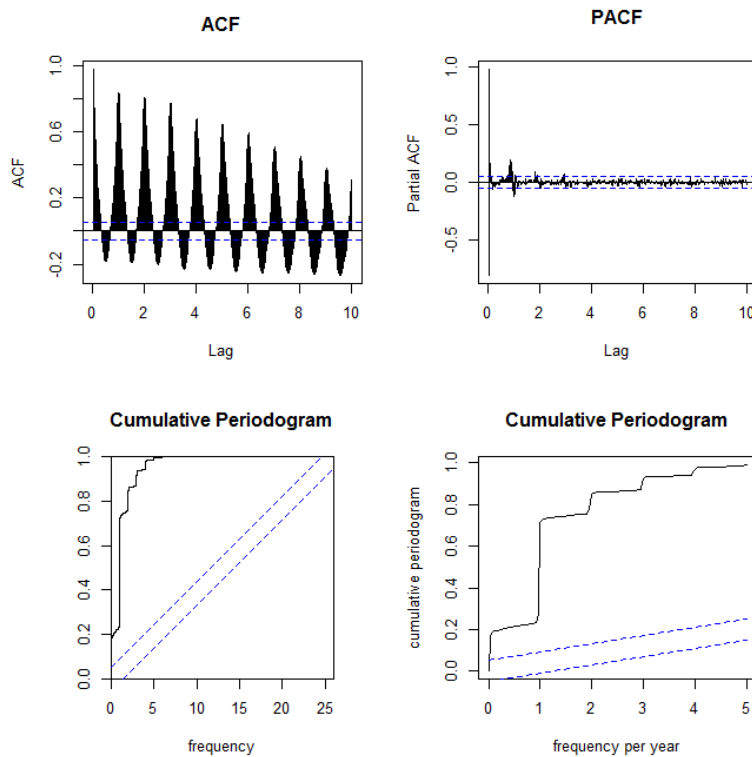


Figure S6.4: Assessment of the seasonal signal in the log transformed time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.3 Method 1: Piecewise Linear Regression

Here we use piecewise linear regression to detrend and seasonally adjust the log transformed time series using function `bfast` (R package `bfast`). Season was specified as "harmonic".

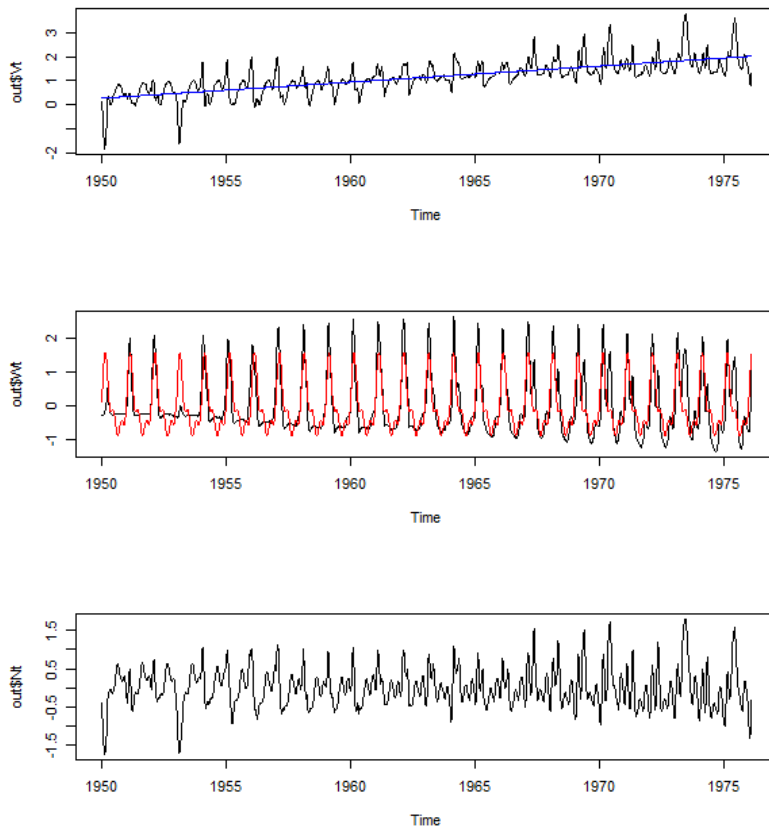


Figure S6.5: Output from the trend and seasonal adjustment by function `bfast` in the R package `bfast` with the overall linear trend (upper panel, blue line), the harmonic seasonal signal (middle panel, red line), and the resulting residual time series (lower panel, black line).

S6.3.1 Seasonal Signals by `auto.arima` on `bfast` Residuals

ARIMA(4,0,4)(0,0,2)[52] with zero mean

Coefficients	ar1	ar2	ar3	ar4	ma1	ma2	ma3	ma4	sma1	sma2
	3.3928	-4.5529	2.8332	-0.6873	-1.2178	0.1264	0.3100	0.0878	0.2116	0.1390
s.e.	0.0473	0.1220	0.1167	0.0407	0.0522	0.0505	0.0468	0.0425	0.0293	0.0288

Σ^2 estimated as 0.003623: log likelihood=1890.37

AIC=-3758.74 AICc=-3758.55 BIC=-3701.38

S6.3.2 Assessing Seasonal Signals by ACF, PACF and Periodogram of bfast Residuals

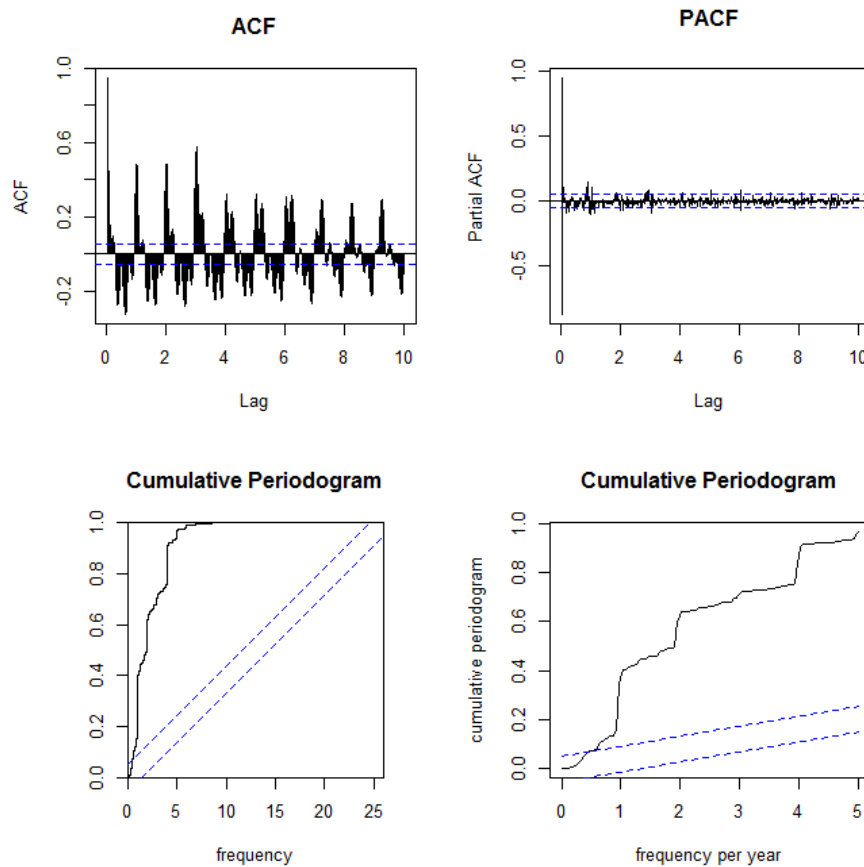


Figure S6.6: Assessment of the seasonal signal in the bfast based residuals time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.3.3 Early Warning Indicators Based on bfast Residuals

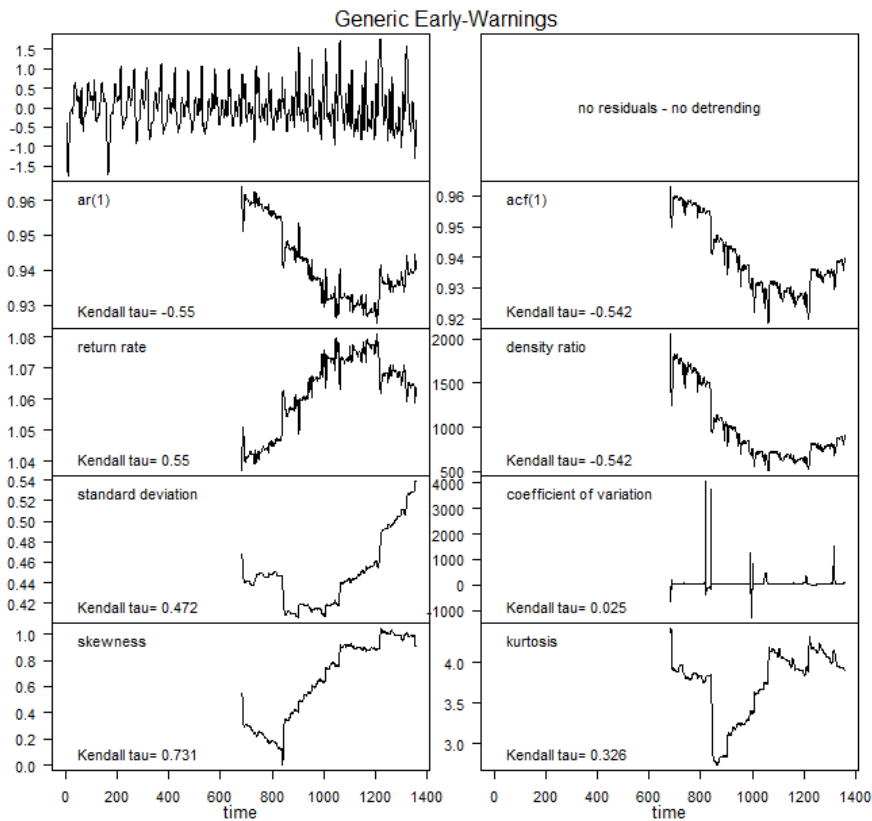


Figure S6.7: Overview of the trend evolution of all EWI metrics on the time series of bfast derived residuals as produced by the R function `generic_ews` (package `earlywarnings`). The left panels, top to bottom show the residuals time series, AR-1 trend, the return rate trend (1-AR-1 coefficient), the SD trend and the SK trend. The right panels, top to bottom show whether additional detrending was employed (none), the autocorrelation at first lag trend, the DR trend (the ratio of low frequencies over high frequencies), the coefficient of variation trend and the kurtosis trend. All metrics are calculated on a rolling window of 50% of the data and their trend evolutions are estimated by the nonparametric Kendall tau correlation. The theoretical expectation is that trends for AR-1, SD, SK, acf(1), DR, coefficient of variation and kurtosis should increase while the trend for return rate should decrease.

S6.4 Method 2: Seasonal Adjustment Using a Stable Seasonal Filter

Here we calculate overall mean values for each week in the year and subsequently subtract the corresponding weekly mean value from the time series. As basis we use the log transformed data in order to stabilize the increase in amplitudes over time.

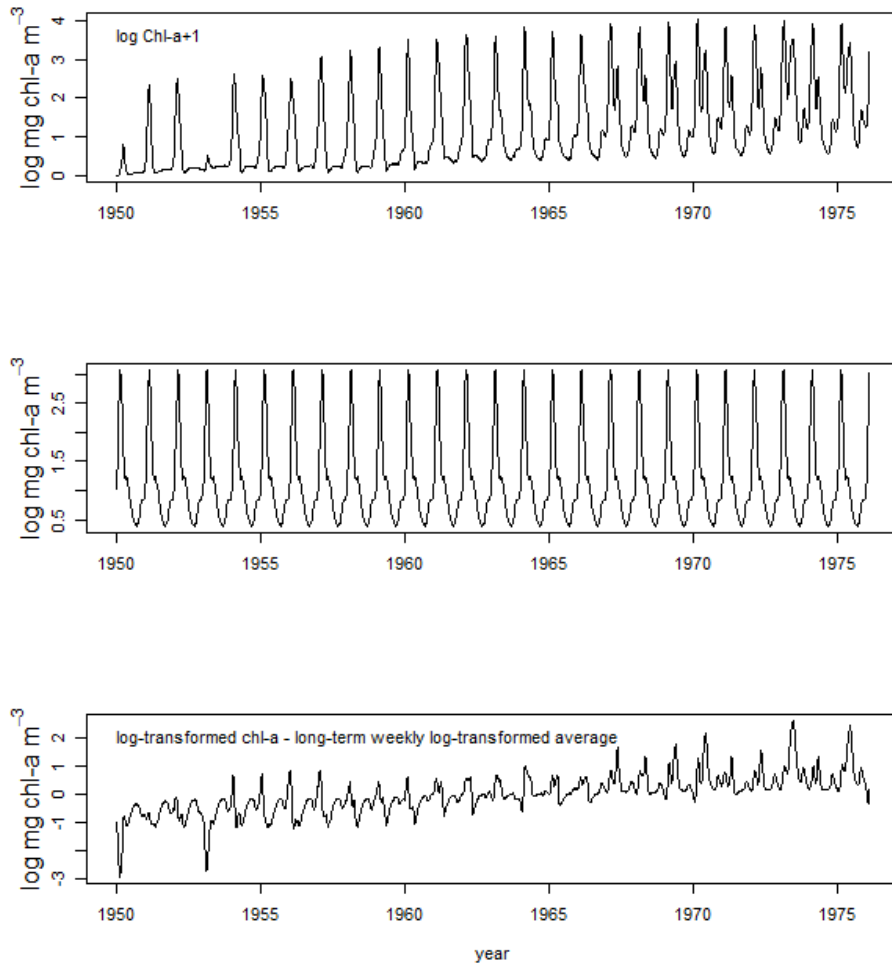


Figure S6.8: The log transformed chlorophyll-a time series serves as basis (upper panel) from which the overall weekly means (middle panel) are deducted resulting in the residuals time series (lower panel).

S6.4.1 Seasonal signals by auto.arima on residuals

ARIMA(2,1,0)(1,0,0)[52]

Coefficients	ar1	ar2	sar1
	1.2329	-0.5330	0.1935
s.e.	0.0243	0.0238	0.0289

Sigma² estimated as 0.003767: log likelihood=1862.52

AIC=-3717.03 AICc=-3717 BIC=-3696.18

S6.4.2 Assessing Seasonal Signals by ACF, PACF and Periodogram of Residuals

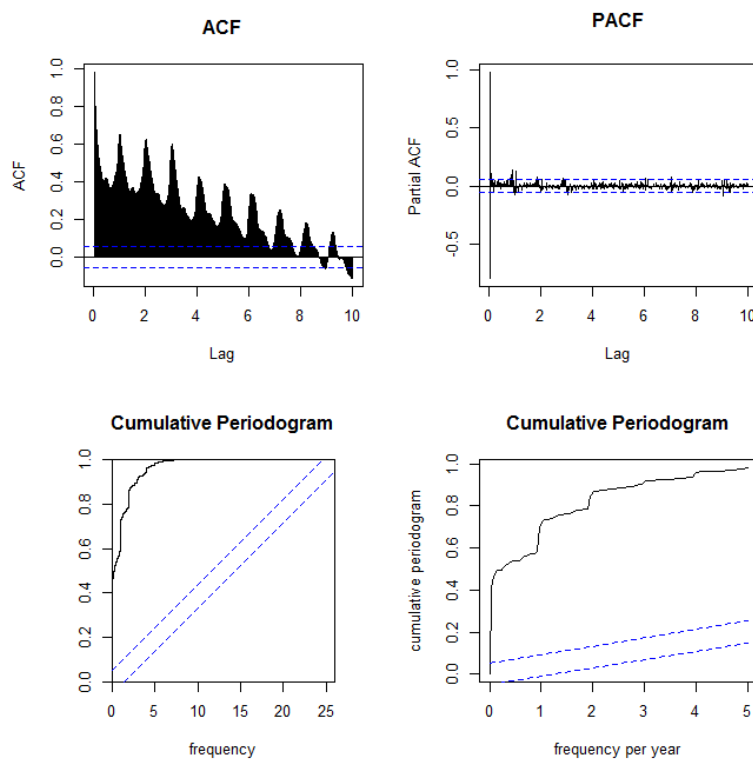


Figure S6.9: Assessment of the seasonal signal in the residuals time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.4.3 Early Warning Indicators Based on Residuals

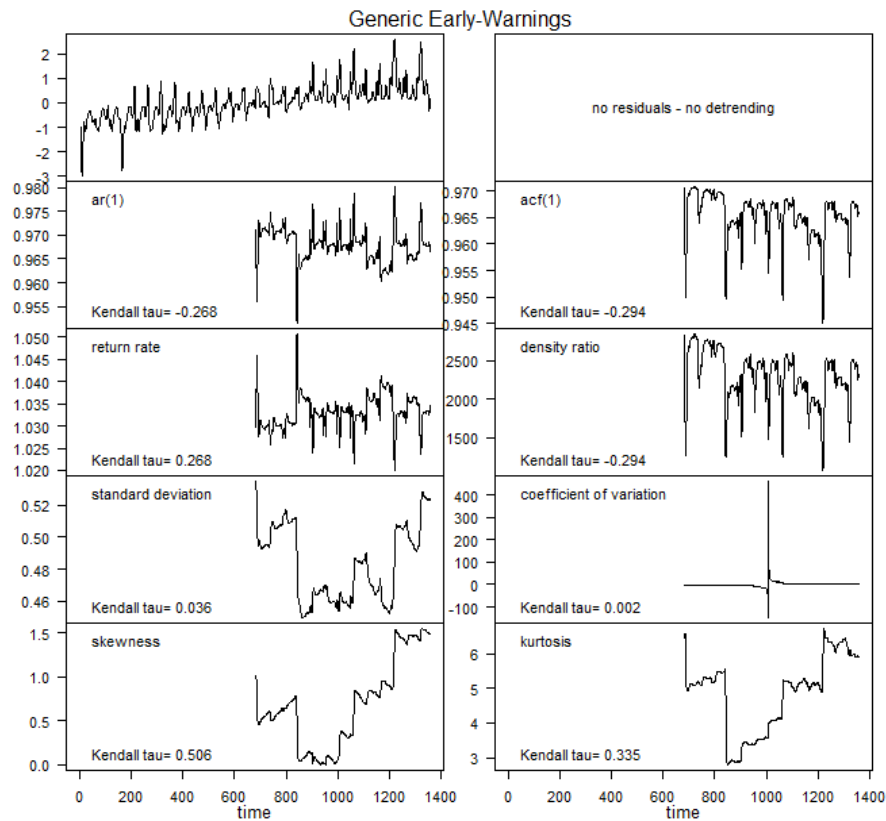


Figure S6.10: Overview of the trend evolution of a EWI metrics on the time series of residuals as produced by the R function `generic_ews` (package `earlywarnings`). The left panels, top to bottom show the residuals time series, $AR-1$ trend, , the return rate trend ($1-AR-1$ coefficient), the SD trend and the SK trend. The right panels, top to bottom show whether additional detrending was employed (none), the autocorrelation at first lag trend, the DR trend (the ratio of low frequencies over high frequencies), the coefficient of variation trend and the kurtosis trend. All metrics are calculated on a rolling window of 50% of the data and their trend evolutions are estimated by the nonparametric Kendall tau correlation. The theoretical expectation is that trends for $AR-1$, SD, SK, $acf(1)$, DR, coefficient of variation and kurtosis should increase while the trend for return rate should decrease.

S6.5 Method 3: Seasonal Decomposition with LOESS

Here we used LOESS based on the log transformed time series, using function “stl” in R with `s.window = 7`

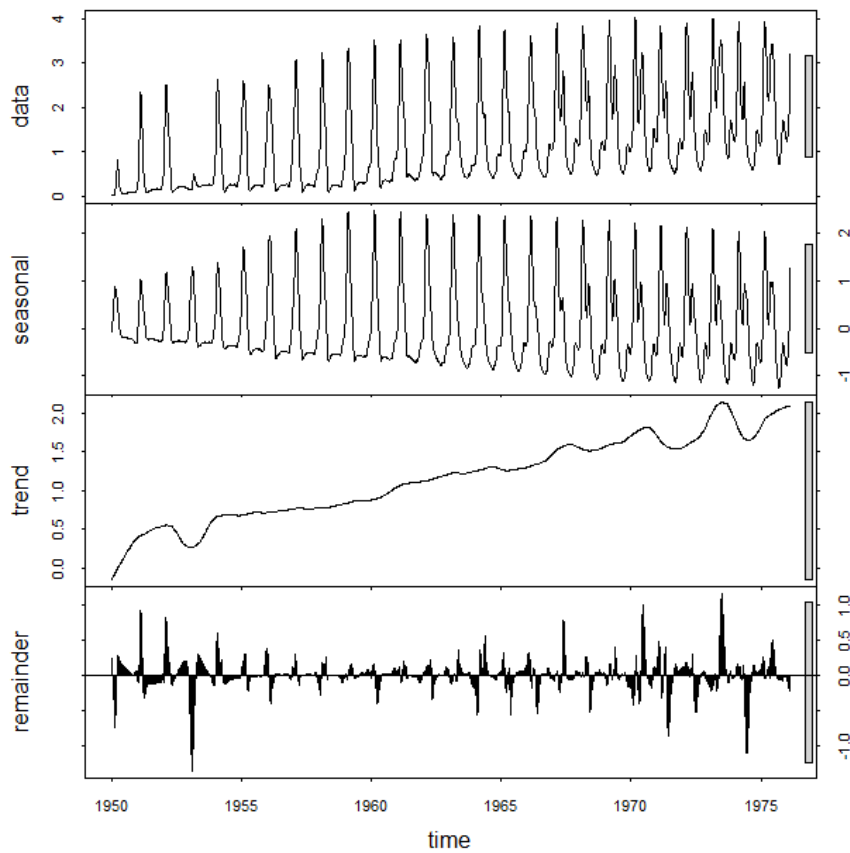


Figure S6.11: The log transformed chlorophyll-a time series serves as basis (upper panel) on which the LOESS function estimated the seasonal (second panel), the trend (third panel) and the so-called remainder components (lowest panel) of the time series.

S6.5.1 Seasonal Signals by auto.arima on Remainder Time Series

ARIMA(2,0,1)(1,0,0)[52] with zero mean

Coefficients	ar1	ar2	ma1	sar1
	1.5919	-0.6988	0.2933	-0.3735
s.e.	0.0233	0.0231	0.0308	0.0258

Sigma² estimated as 0.001606: log likelihood=2439.56

AIC=-4869.13 AICc=-4869.08 BIC=-4843.05

S6.5.2 Assessing Seasonal Signals by ACF, PACF and Periodogram of Remainder Time Series

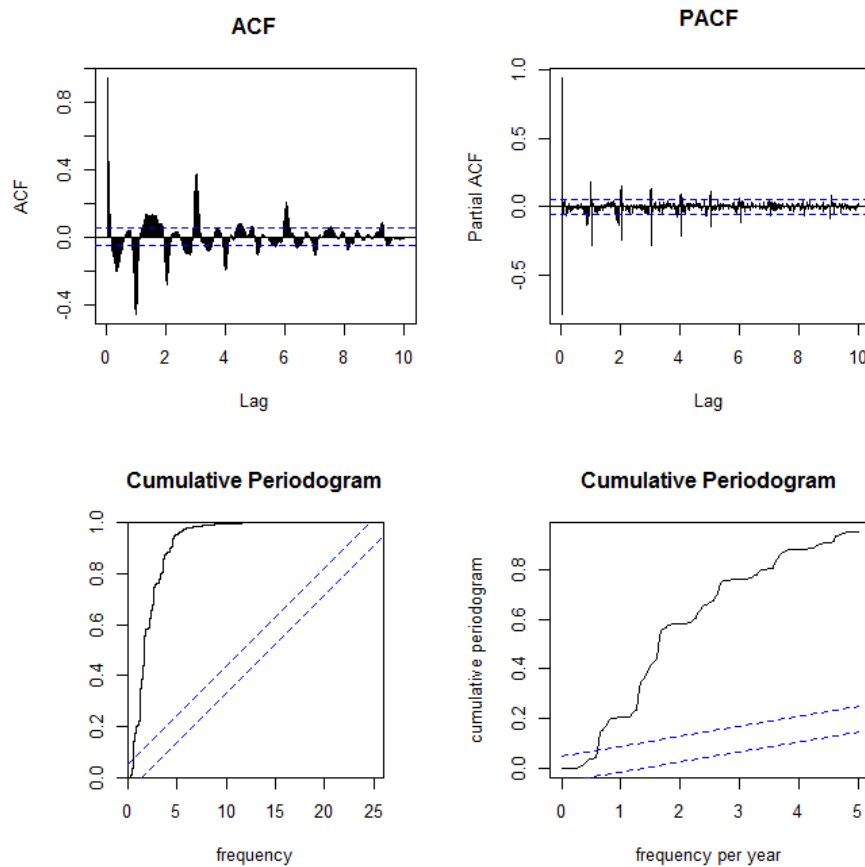


Figure S6.12: Assessment of the seasonal signal in the reminder time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.5.3 Early Warning Indicators Based on Remainder

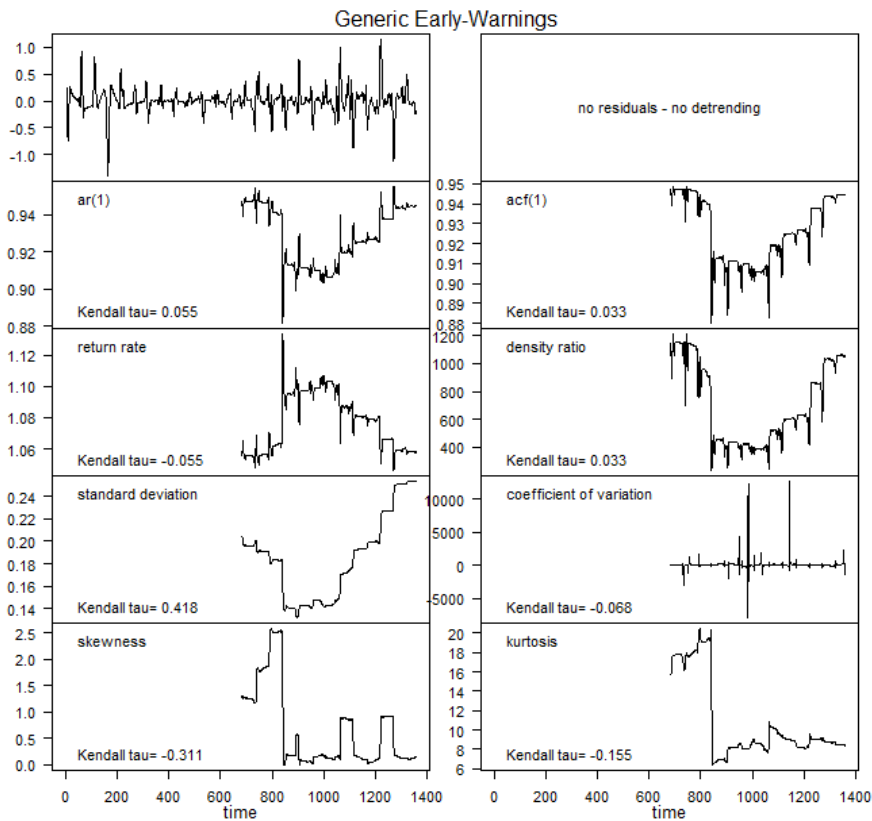


Figure S6.13: Overview of the trend evolution of a EWI metrics on the time series of remainder as produced by the R function `generic_ews` (package `earlywarnings`). The left panels, top to bottom show the residuals time series, AR-1 trend, the return rate trend (1-AR-1 coefficient), the SD trend and the SK trend. The right panels, top to bottom show whether additional detrending was employed (none), the autocorrelation at first lag trend, the DR trend (the ratio of low frequencies over high frequencies), the coefficient of variation trend and the kurtosis trend. All metrics are calculated on a rolling window of 50% of the data and their trend evolutions are estimated by the nonparametric Kendall tau correlation. The theoretical expectation is that trends for AR-1, SD, SK, acf(1), DR, coefficient of variation and kurtosis should increase while the trend for return rate should decrease.

S6.6 Method 4: Seasonal Decomposition by Moving Averages

Here we use the function “decompose” in R to detrend and seasonally adjust the original time series. To allow the decomposition method to deal dynamically with the increase in seasonal amplitude, we chose the function type as "multiplicative".

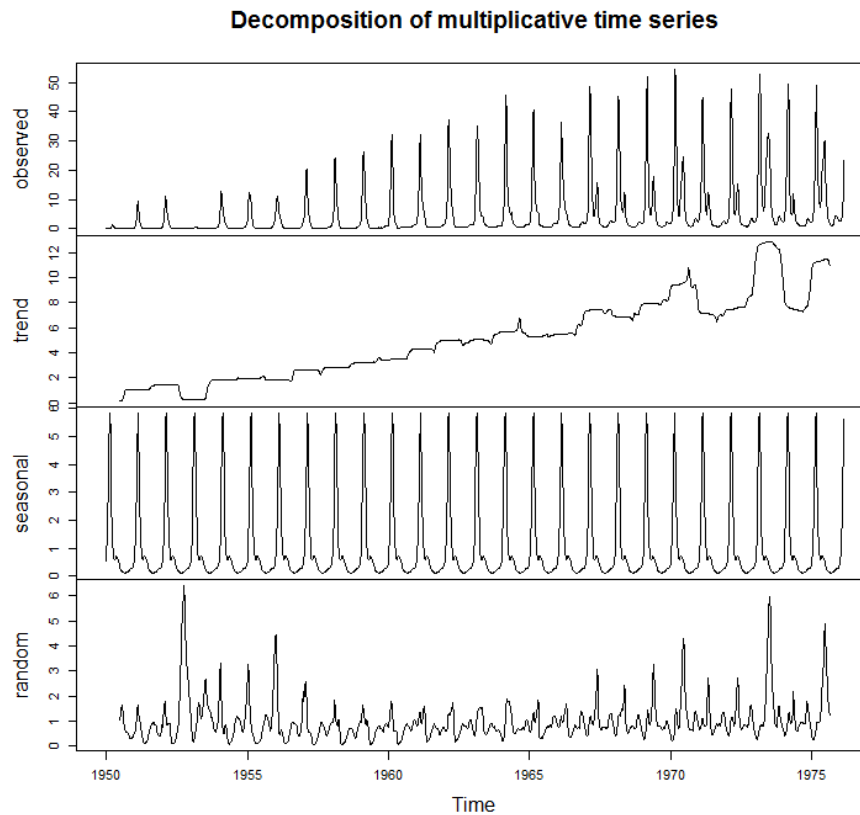


Figure S6.14: The original chlorophyll-a time series serves as basis (upper panel) on which the decompose function estimated the trend (second panel), the seasonal (third panel) and the so-called random components (lowest panel) of the time series.

S6.6.1 Seasonal Signals by auto.arima on the “Random” Time Series

ARIMA(2,0,1) with non-zero mean

Coefficients	ar1	ar2	ma1	intercept
	1.6205	-0.6898	0.3783	0.9954
s.e.	0.0235	0.0234	0.0296	0.0669

Σ^2 estimated as 0.01494: log likelihood=892.21

AIC=-1774.41 AICc=-1774.37 BIC=-1748.54

S6.6.2 Assessing Seasonal Signals by ACF, PACF and Periodogram of the Random Time Series

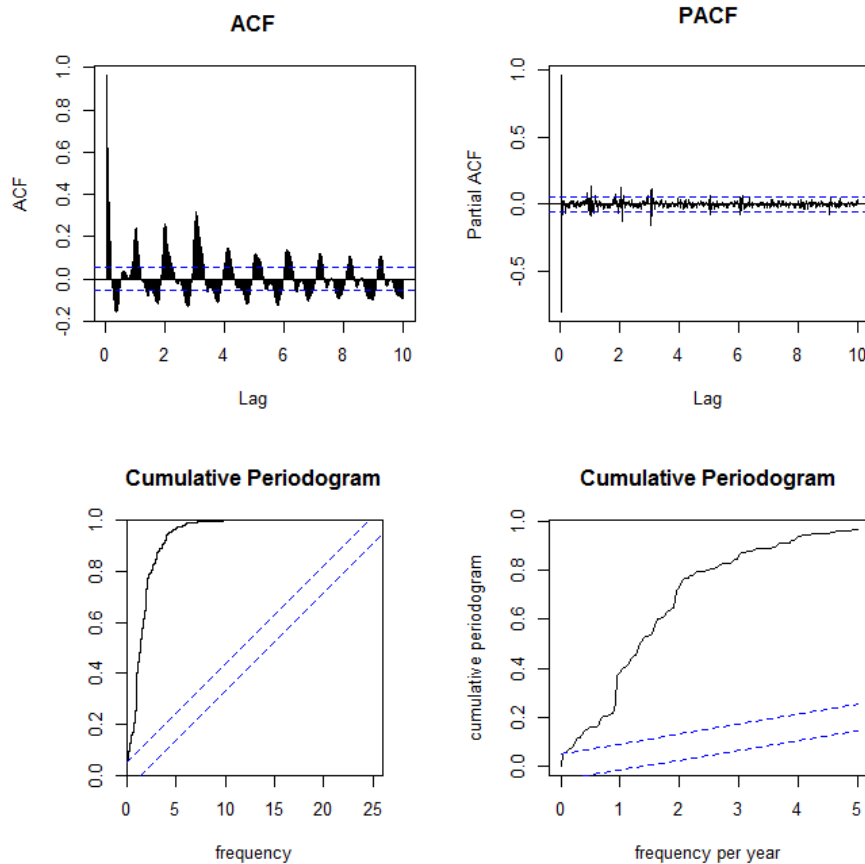


Figure S6.15: Assessment of the seasonal signal in the random time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.6.3 Early Warning Indicators Based on Random Time Series

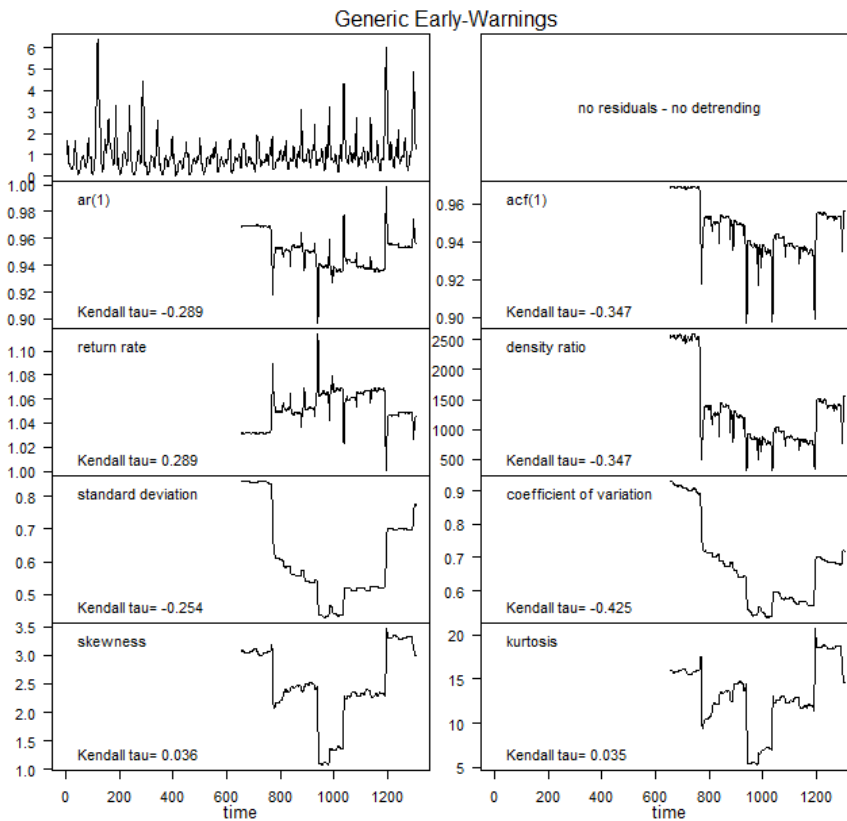


Figure S6.16: Overview of the trend evolution of a EWI metrics on the time series of random component as produced by the R function `generic_ews` (package `earlywarnings`). The left panels, top to bottom show the residuals time series, AR-1 trend, the return rate trend (1-AR-1 coefficient), the SD trend and the SK trend. The right panels, top to bottom show whether additional detrending was employed (none), the autocorrelation at first lag trend, the DR trend (the ratio of low frequencies over high frequencies), the coefficient of variation trend and the kurtosis trend. All metrics are calculated on a rolling window of 50% of the data and their trend evolutions are estimated by the nonparametric Kendall tau correlation. The theoretical expectation is that trends for AR-1, SD, SK, acf(1), DR, coefficient of variation and kurtosis should increase while the trend for return rate should decrease.

S6.7 Method 5: First Differencing

Here we used first differencing to seasonally adjust the log transformed chlorophyll-a time series, using the “diff” function in R.

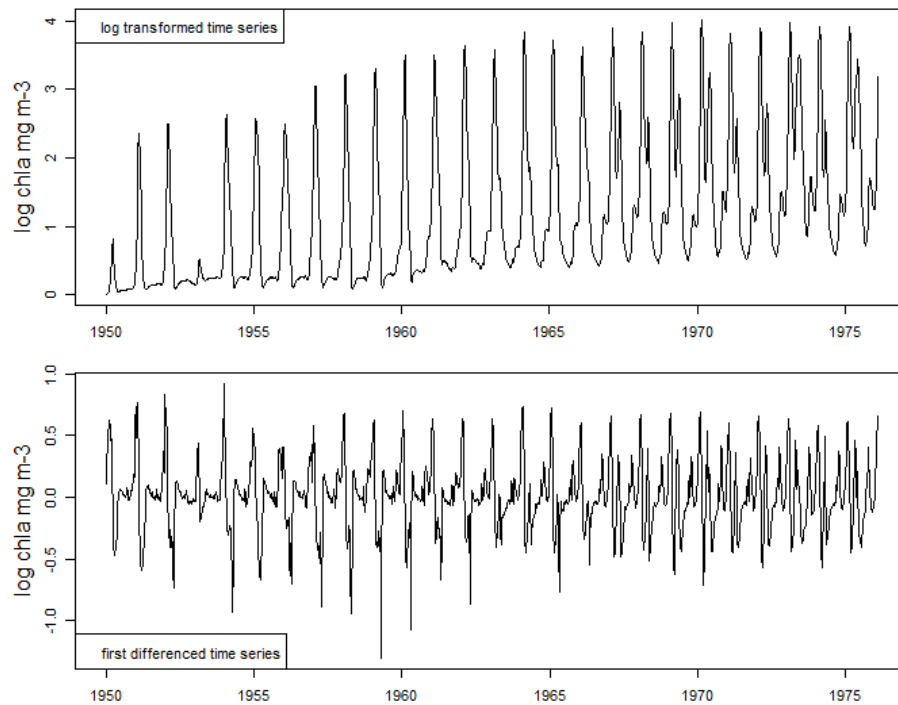


Figure S6.17: The log transformed chlorophyll-a time series serves as basis (upper panel) on which the first differencing is performed by function “diff” in R resulting in the first differenced time series (lower panel).

S6.7.1 Seasonal Signals by auto.arima on Differenced Time Series

ARIMA(2,0,1)(1,0,0)[52] with zero mean

Coefficients	ar1	ar2	ma1	sar1
	1.3795	-0.5706	-0.3129	0.4677
s.e.	0.1067	0.0845	0.1342	0.0280

σ^2 estimated as 0.01099: log likelihood=1130.98

AIC=-2251.96 AICc=-2251.91 BIC=-2225.89

S6.7.2 Assessing Seasonal Signals by ACF, PACF and Periodogram of the Differenced Time Series

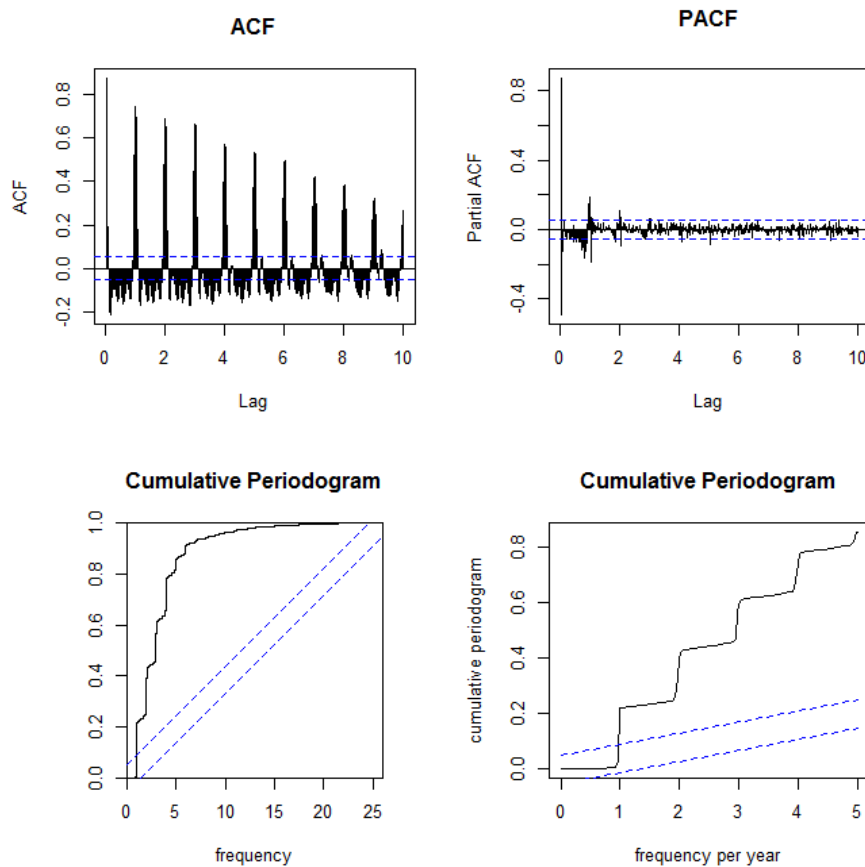


Figure S6.18: Assessment of the seasonal signal in the differenced time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.7.3 Early Warning Indicators Based on the Differenced Time Series

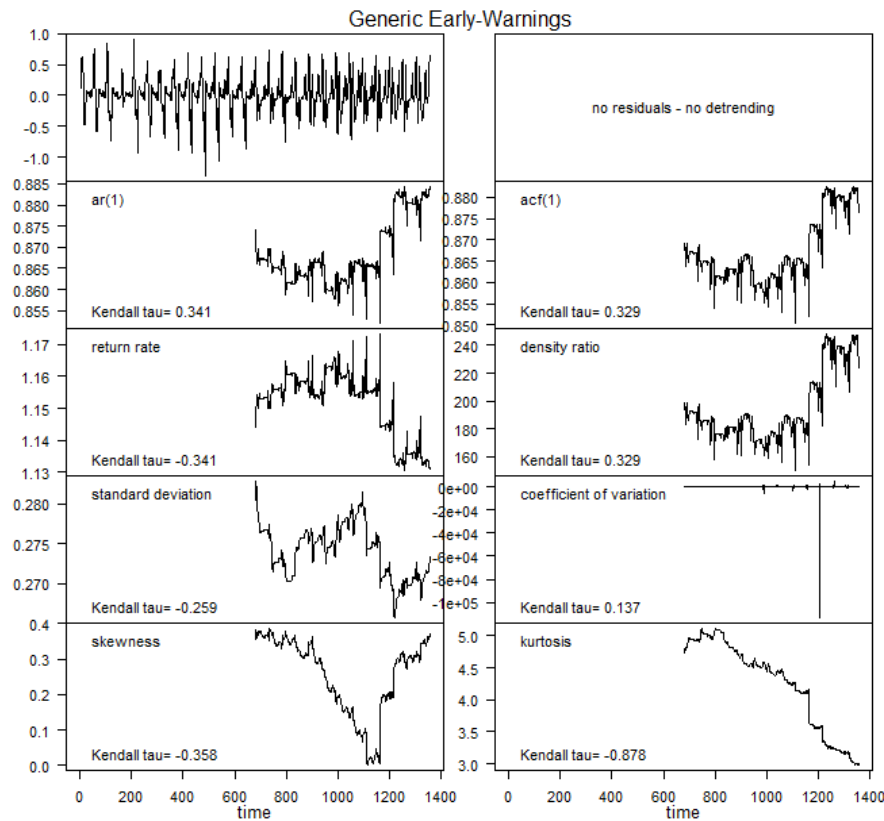


Figure S6.19: Overview of the trend evolution of a EWI metrics on the differenced time series as produced by the R function `generic_ews` (package `earlywarnings`). The left panels, top to bottom show the residuals time series, $AR(1)$ trend, the return rate trend ($1-AR(1)$ coefficient), the SD trend and the SK trend. The right panels, top to bottom show whether additional detrending was employed (none), the autocorrelation at first lag trend, the DR trend (the ratio of low frequencies over high frequencies), the coefficient of variation trend and the kurtosis trend. All metrics are calculated on a rolling window of 50% of the data and their trend evolutions are estimated by the nonparametric Kendall tau correlation. The theoretical expectation is that trends for $AR(1)$, SD, SK, $acf(1)$, DR, coefficient of variation and kurtosis should increase while the trend for return rate should decrease.

S6.8 Method 6: Gaussian Filtering Using Bandwidth Fixed to One Year of Data

Here we use Gaussian filtering (function `ksmooth` in R) with a bandwidth fixed to the percentage of data that represents one year in order to seasonally adjust and detrend the log transformed time series.

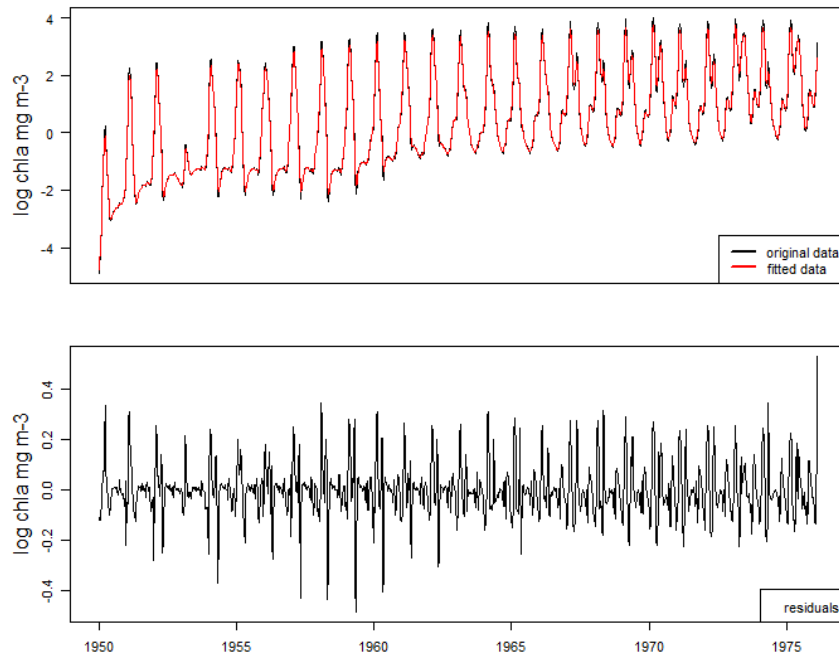


Figure S6.20: The log transformed chlorophyll-a time series serves as basis (upper panel, black line) on which the Gaussian filtering is performed (function “`ksmooth`” in R, red line) resulting in the residuals time series (lower panel).

S6.8.1 Seasonal Signals by `auto.arima` on Residuals Time Series

ARIMA(3,0,1)(1,0,0)[52] with zero mean

Coefficients	ar1	ar2	ar3	ma1	sar1
	0.3997	0.0486	-0.1905	0.5728	0.4486
s.e.	0.2108	0.2048	0.1029	0.2056	0.0281

σ^2 estimated as 0.003012: log likelihood=2012.37

AIC=-4012.74 AICc=-4012.68 BIC=-3981.45

S6.8.2 Assessing Seasonal Signals by acf, pacf and Periodogram of the Residuals Time Series

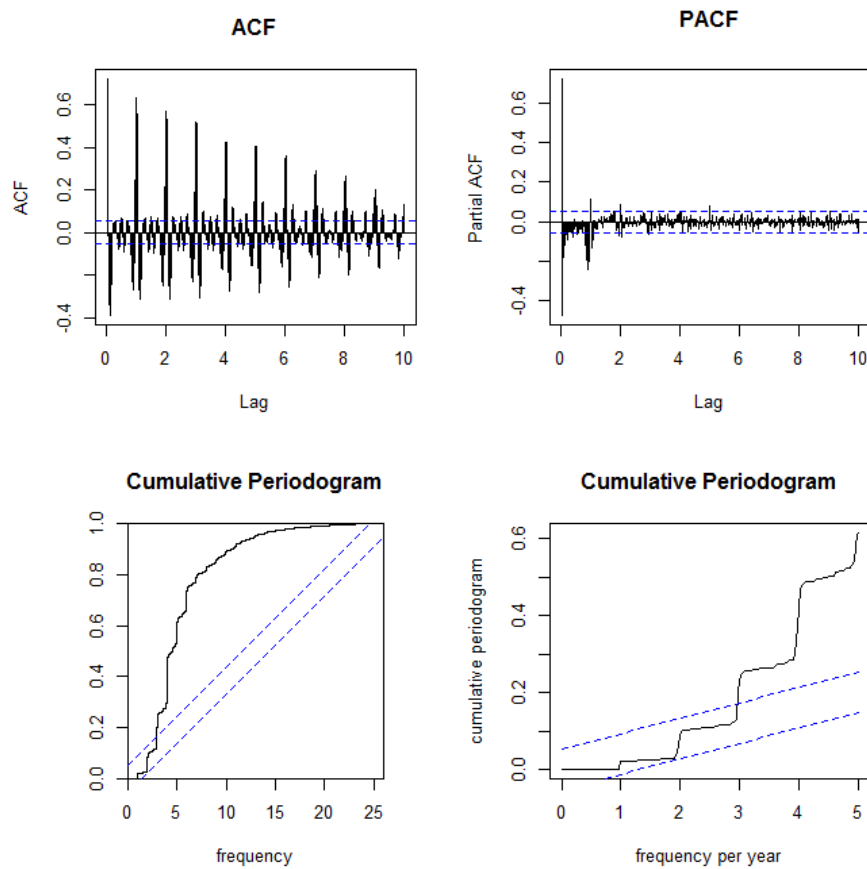


Figure S6.21: Assessment of the seasonal signal in the residuals time series using ACF (upper left panel), PACF (upper right panel), and the cumulative periodogram (lower left panel). For improved clarity, a magnified section of the cumulative periodogram is included, showing the cumulative energy up to frequency 5 (lower right panel). Blue lines represent the upper and lower 95% confidence interval for white noise processes.

S6.8.3 Early Warning Indicators Based on Residual Time Series

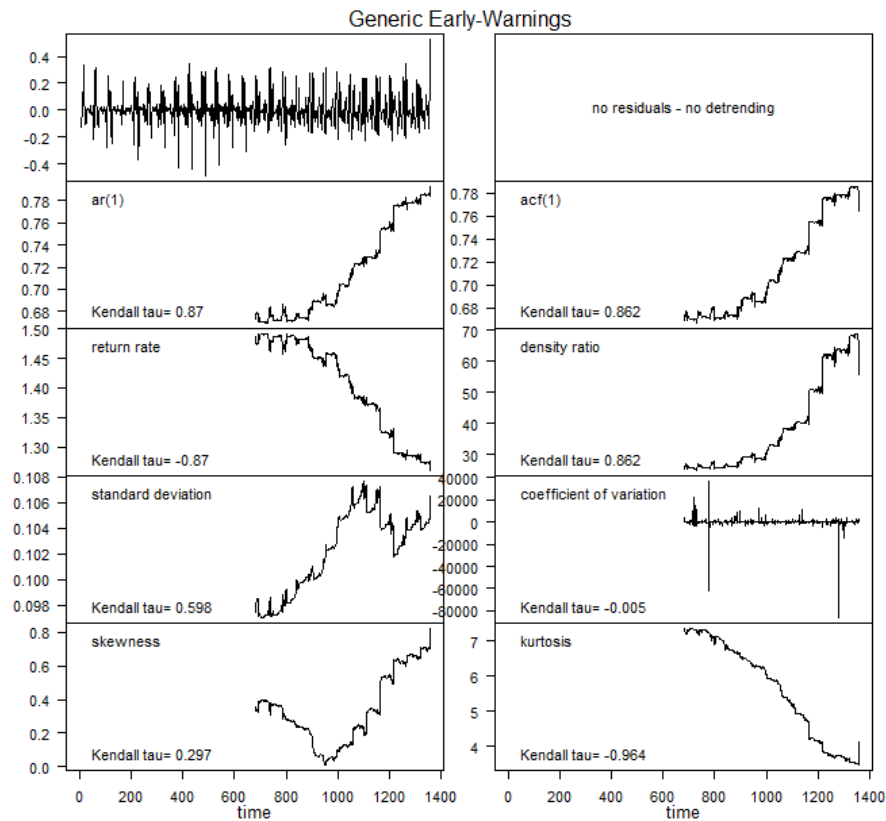


Figure S6.22: Overview of the trend evolution of a EWI metrics on the residuals time series as produced by the R function `generic_ews` (package `earlywarnings`). The left panels, top to bottom show the residuals time series, $AR(1)$ trend, the return rate trend ($1-AR(1)$ coefficient), the SD trend and the SK trend. The right panels, top to bottom show whether additional detrending was employed (none), the autocorrelation at first lag trend, the DR trend (the ratio of low frequencies over high frequencies), the coefficient of variation trend and the kurtosis trend. All metrics are calculated on a rolling window of 50% of the data and their trend evolutions are estimated by the nonparametric Kendall tau correlation. The theoretical expectation is that trends for $AR(1)$, SD, SK, $acf(1)$, DR, coefficient of variation and kurtosis should increase while the trend for return rate should decrease.

S6.9 Comparison of all methods

Here we plotted all cumulative periodogram lines into one plot for easier comparison. Most methods have an even higher cumulative energy at frequency 2 (hence seasonally recurrent events) than the original time series, only first differencing (Method 5) and Gaussian filtering (Method 6) perform better. Gaussian filtering seems to be able to remove seasonal signals best as the cumulative energy at frequency 2 remains within the confidence limits for a white noise process (Figure S6.23). Moreover, comparing the EWI trends based on first differencing and Gaussian filtering suggests that first differencing does change the outcome of EWI analysis (Figure S6.19) while Gaussian filtering seems to retain the expected characteristics of the EWI trends (Figure S6.22). Based on these preliminary method tests, we chose to use Gaussian filtering for the seasonal adjustment of the time series in the main manuscript.

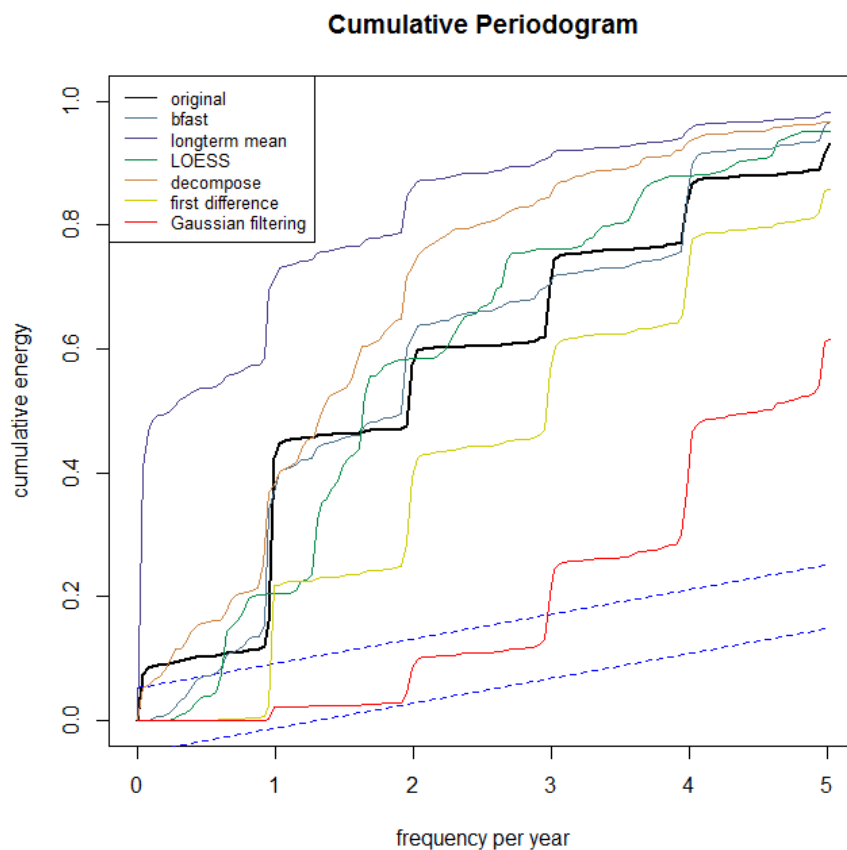


Figure S6.23: Assessment of the seasonal signal in the residuals time series a magnified section of the cumulative periodogram, showing the cumulative energy up to frequency 5. Blue lines represent the upper and lower 95% confidence interval for white noise processes.

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